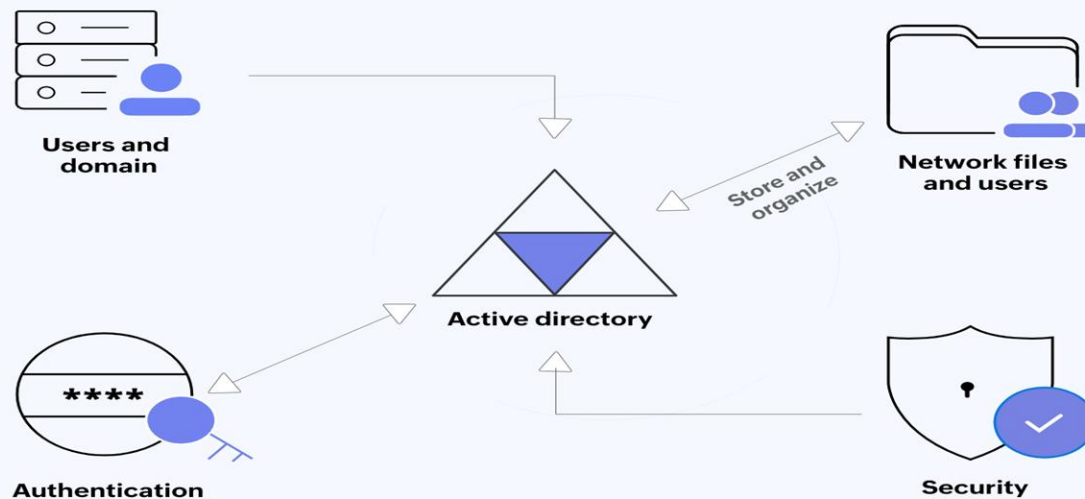




Module 02

Active Directory



1. Introduction to Active Directory

- 1.1. Domain Controller
- 1.2. Active Directory

2. Prepare before Installing the AD DS role

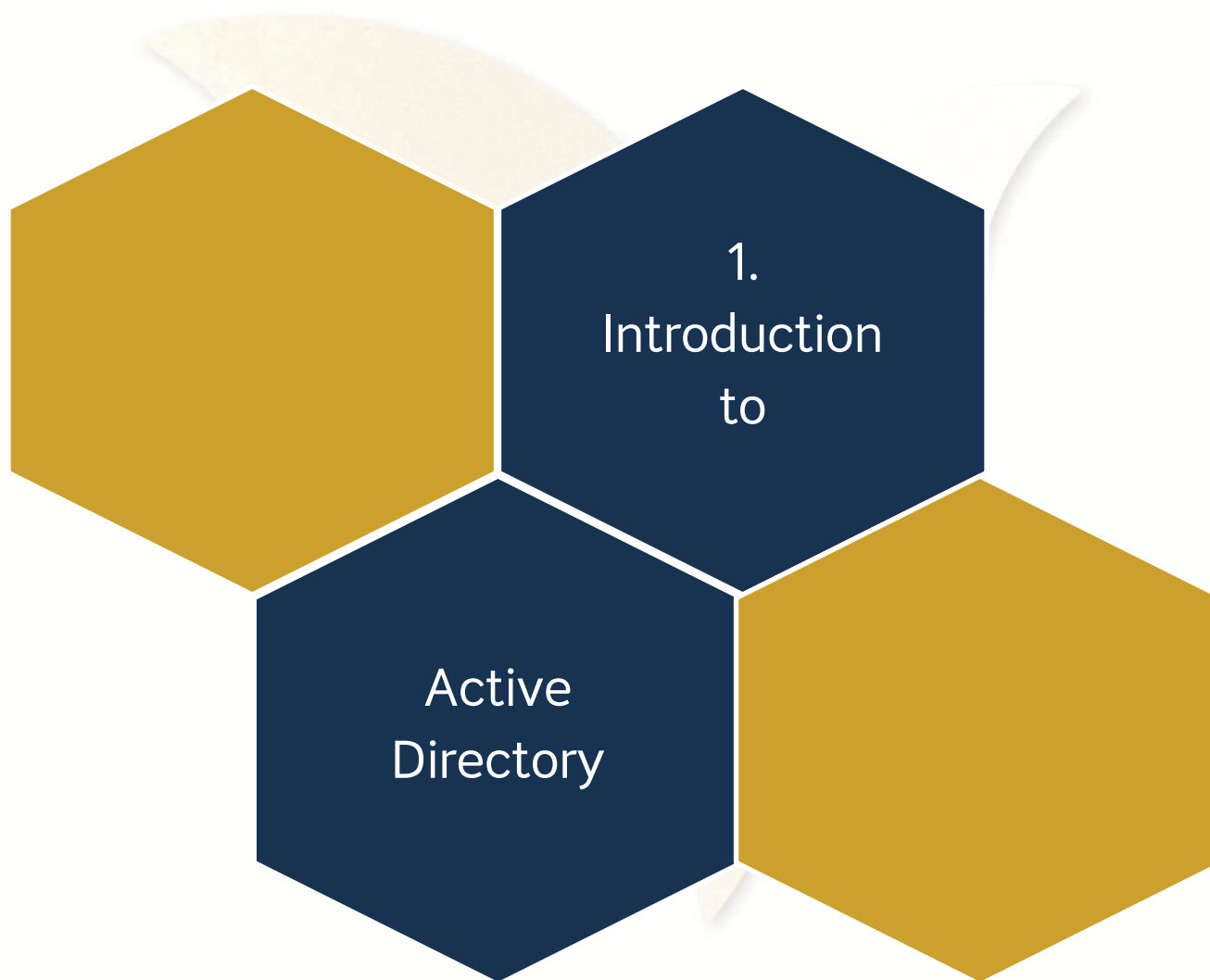
- 2.1. Server Requirements
- 2.2. Change Hostname
- 2.3. Set IP Address

3. Install the AD DS role

- 3.1. Installation
- 3.2. Create a User
- 3.3. Join a Computer to a Domain

4. Backup Active Directory

- 4.1. Install Windows Backup Feature
- 4.2. Backup Once AD
- 4.3. Restore AD



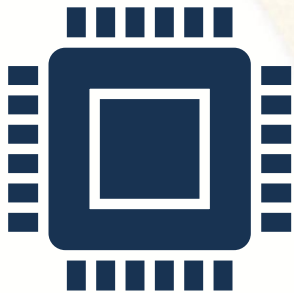
1.1. Domain Controller



A **Domain Controller**, commonly referred to as a **DC**, is simply a server that is hosting **Active Directory**. It is a central point of contact, a central "hub" so to speak, that is accessed prior to almost any communication that takes place between a client and server in your network.

Perhaps the easiest way to describe it is as a storage container for all identification that happens on the **network**. **Usernames, passwords, computer accounts, groups of computers, servers, groups and collections of servers, security policies, file replication services,** and many more things are stored within and managed by DCs.

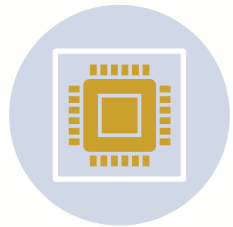
1.1. Domain Controller (Cont.)



Domain Controller is a server that monitors and manages domains such as providing authentication and authorization. Authentication is the setting of permissions for clients to log on or not. Authorization is the setting of permissions for clients to log on and what they have the right to do.

A domain is a collection of devices such as Servers, Computers, Printers, Routers, Users, ... to connect to a server-based network of an organization or a company. We can say that one network is equal to one domain and in each domain, there may be one or more Domain Controllers. Each domain must have a logical name, and the domain name should always be patterned after the organization name, for example: ***makara.com***

1.2. Active Directory



To become a server that can manage and provide services to clients, we need to make it a Domain Controller first. So, if we want to set up a **Domain Controller**, we need to set up Active Directory first, then it provides us with a Domain Controller as well.



Active Directory is a database used to centrally manage and store all information from a network or from an organization's domain. Alternatively, it is a database that stores information from network objects, such as computers, printers. , Users, Applications, Security Policy, database,



2. Prepare before Installing the AD DS role

2.1. Server Requirements



Set a static IP address

192.168.1.2/24



Set a good hostname

WIN-SERVER



Set a domain name

makara.com

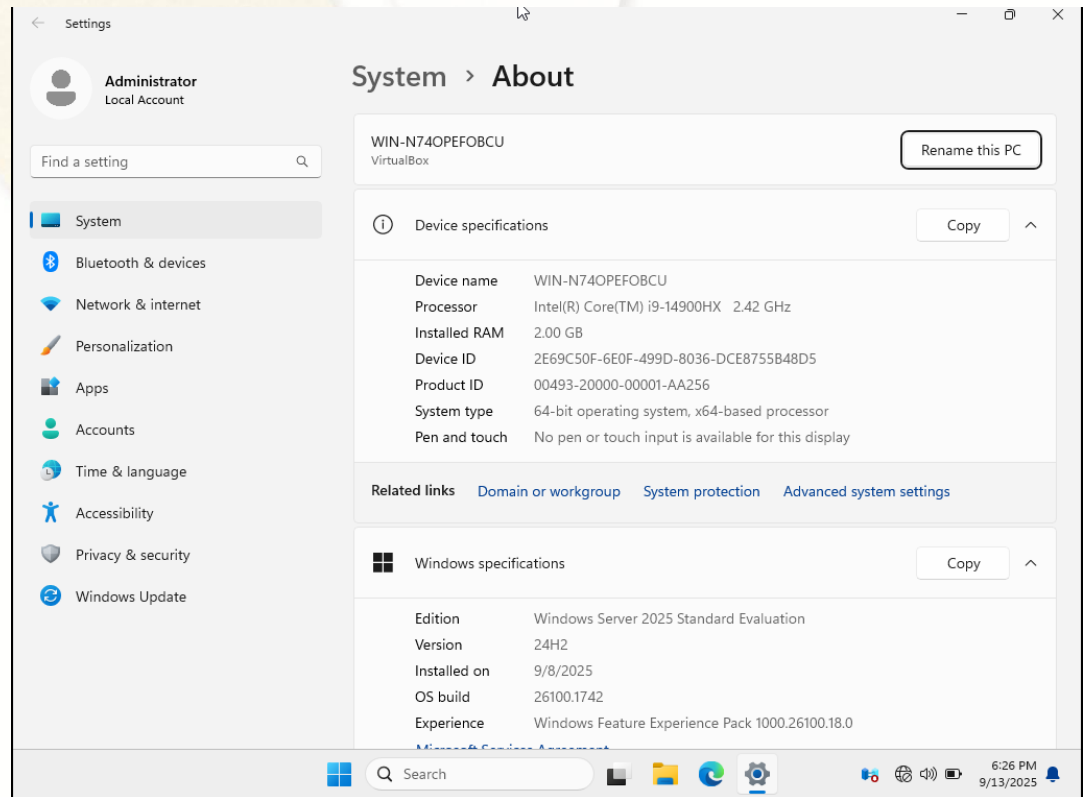
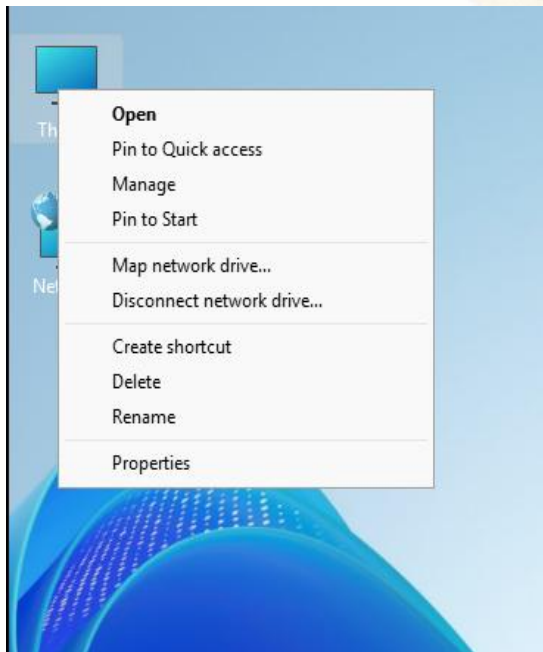


Set a DNS server address

192.168.1.2

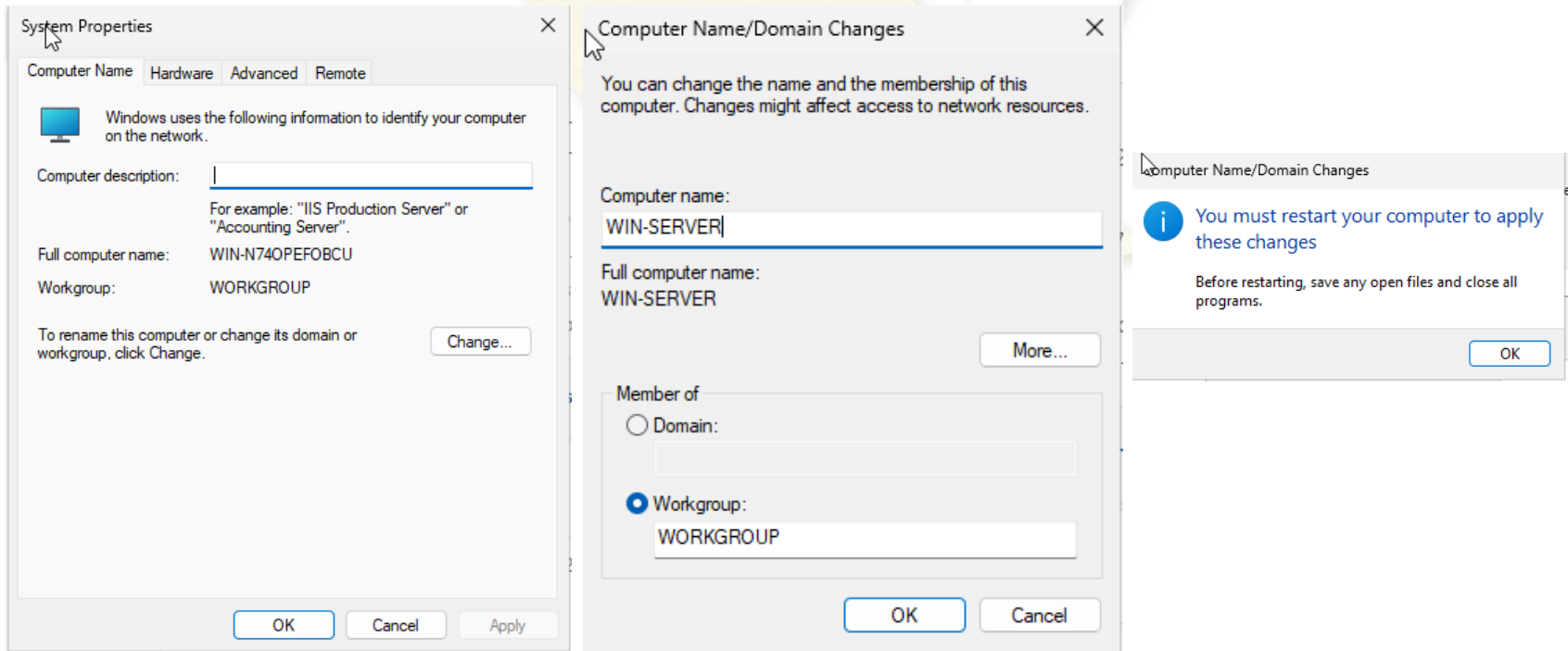
2.2. Change Hostname

- Step 1: Right click on **This PC** > Choose **Properties** > Click **Domain or workgroup** link



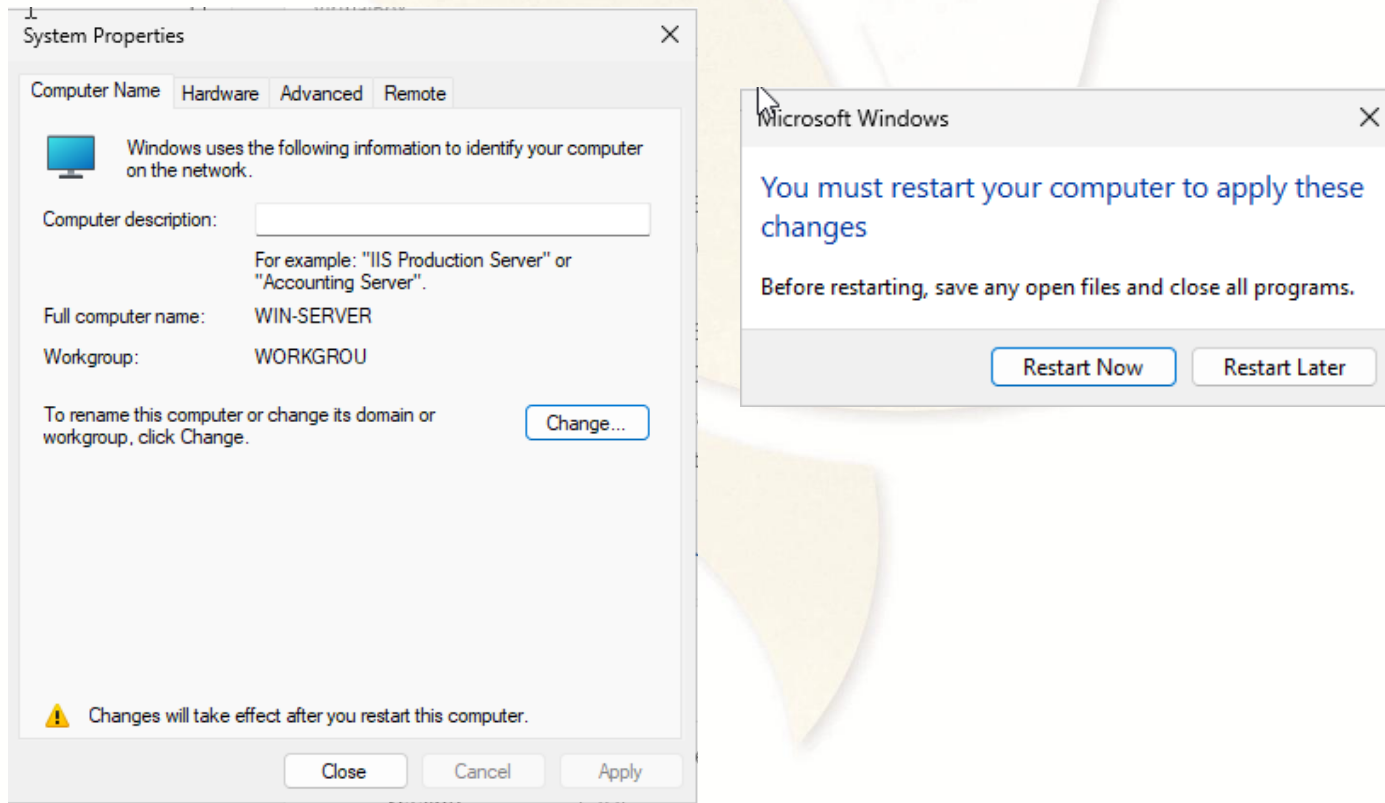
2.2. Change Hostname (Cont.)

- Step 2: Click **Change** button > In **Computer name:** Enter new hostname (**WIN-SERVER**) > Click **OK** > Click **OK**



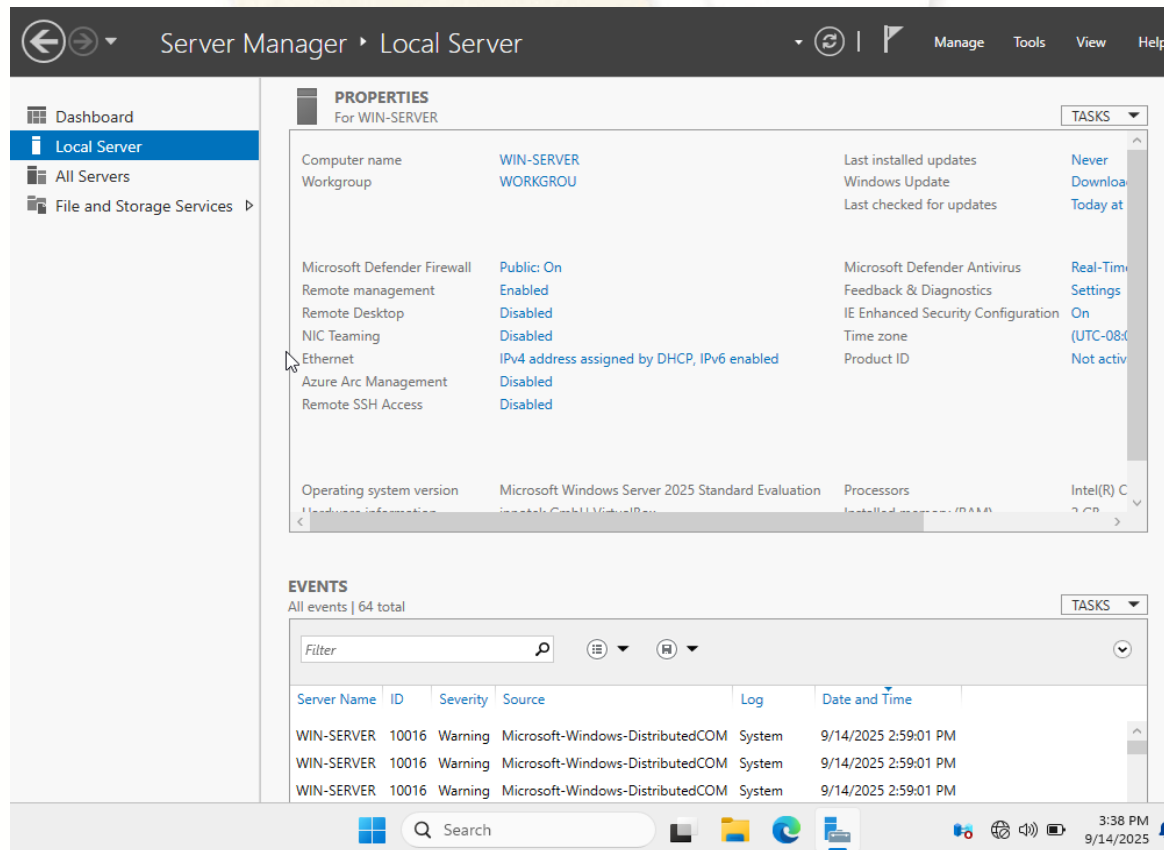
2.2 Change Hostname (Cont.)

- Step 3: Click **Close** > Click **Restart Now** button



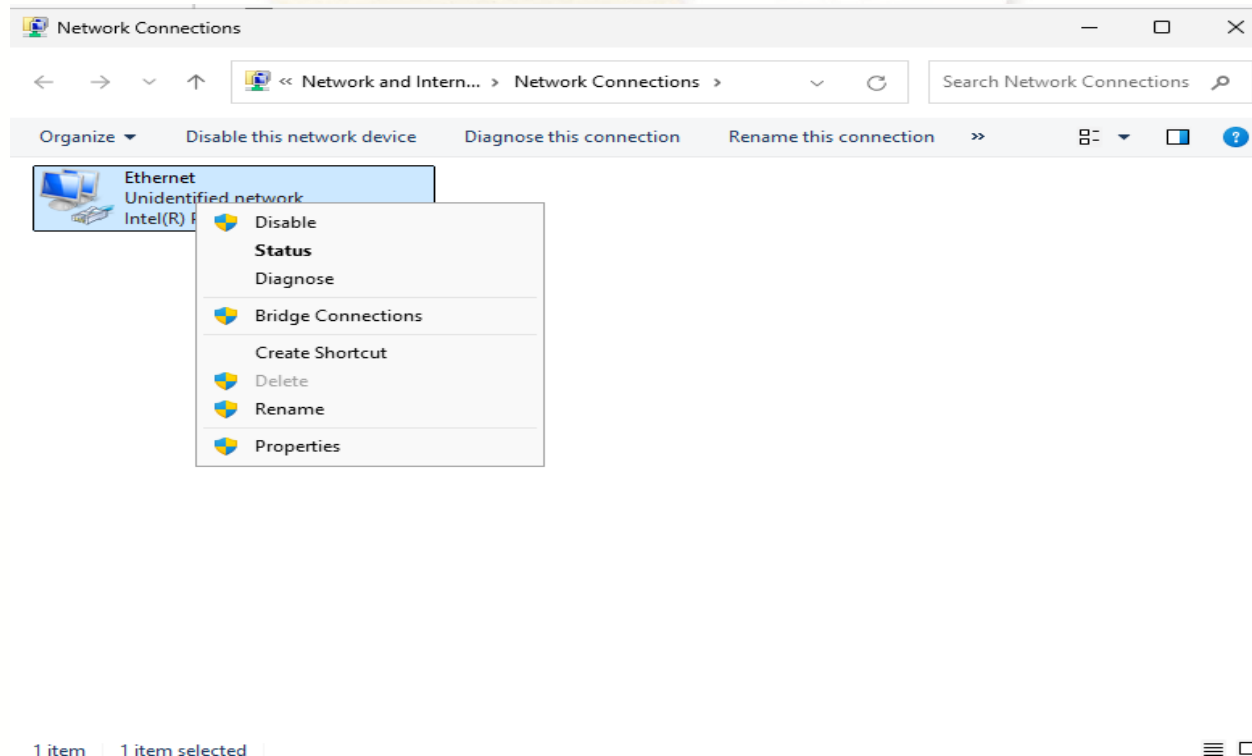
2.3. Set IP Address

- Step 1: Click on **Local Server** in **Server Manager** icon > Choose on Ethernet **IPv4 address assigned by DHCP, IPv6 enabled**



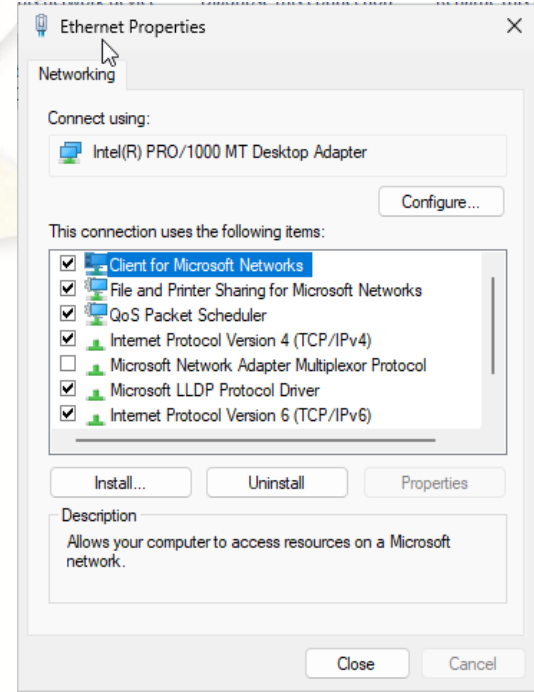
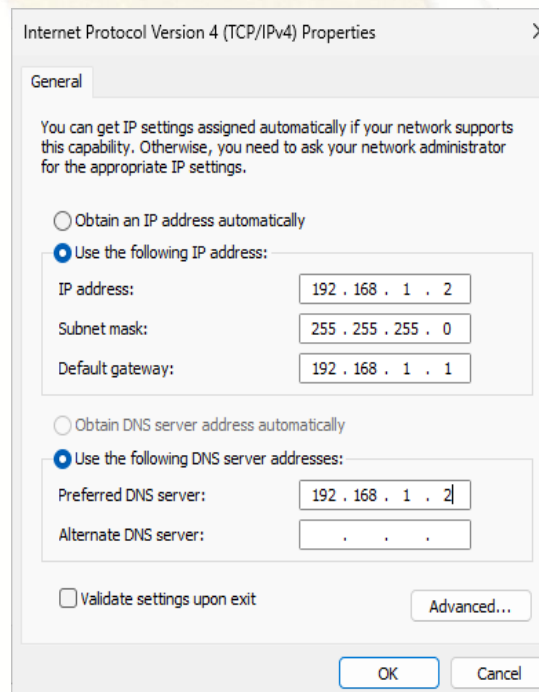
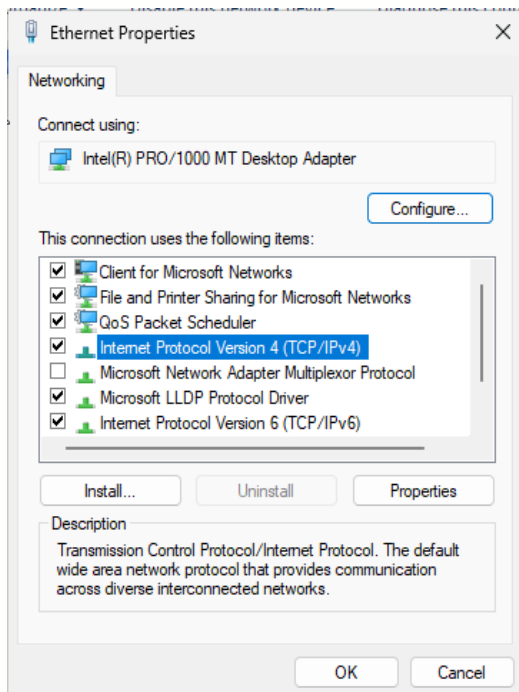
2.3. Set IP Address (Cont.)

- Step 2: Right click on **Network Adapter** (Ethernet) > Choose **Properties**



2.3. Set IP Address (Cont.)

- Step 3: Select **Internet Protocol Version 4 (TCP/IPv4)**
 - > Click **Properties** button > Enter **IP address**, **Subnet mask**, **Default gateway** and **Preferred DNS server** > Click **OK** > Click **Close**

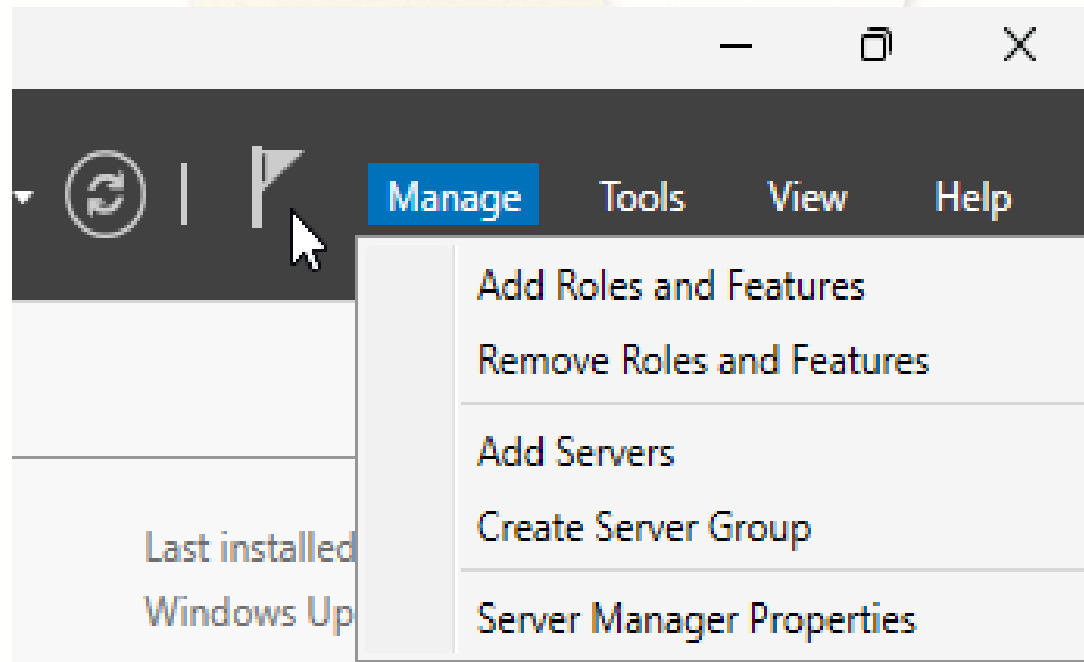




3. Install the AD DS role

3.1. Installation

- Step 1: Click **Manage** menu > Click **Add Roles and Features**



3.1. Installation (Cont.)

- Step 2: On **Before you begin** window, click **Next** button > On **Select installation type** windows, click **Next** button >

The screenshot displays the 'Add Roles and Features Wizard' in two overlapping windows. The foreground window is titled 'Before you begin' and contains the following content:

- Before You Begin** (selected in the left sidebar)
- Installation Type
- Server Selection
- Server Roles
- Features
- Confirmation
- Results

Main text in the 'Before you begin' window:

This wizard helps you install roles, role services, or features. You determine features to install based on the computing needs of your organization hosting a website.

To remove roles, role services, or features:
[Start the Remove Roles and Features Wizard](#)

Before you continue, verify that the following tasks have been completed:

- The Administrator account has a strong password
- Network settings, such as static IP addresses, are configured
- The most current security updates from Windows Update are installed

If you must verify that any of the preceding prerequisites have been completed, complete the steps, and then run the wizard again.

To continue, click Next.

☐ Skip this page by default

Navigation buttons: < Previous, Next >

The background window is titled 'Select installation type' and contains the following content:

- Before You Begin**
- Installation Type** (selected in the left sidebar)
- Server Selection
- Server Roles
- Features
- Confirmation
- Results

Main text in the 'Select installation type' window:

Select the installation type. You can install roles and features on a running physical computer or virtual machine, or on an offline virtual hard disk (VHD).

- ☒ **Role-based or feature-based installation**
Configure a single server by adding roles, role services, and features.
- ☐ **Remote Desktop Services installation**
Install required role services for Virtual Desktop Infrastructure (VDI) to create a virtual machine-based or session-based desktop deployment.

Navigation buttons: < Previous, Next >, Install, Cancel

3.1. Installation (Cont.)

- Step 3: On Select destination server, click Next button > On Select server roles, check Active Directory Domain Services

The image displays two screenshots of the 'Add Roles and Features Wizard' in Windows Server 2012 R2.

Left Screenshot: Select destination server

The wizard is titled 'Add Roles and Features Wizard'. The left sidebar shows the progression: Before You Begin, Installation Type, **Server Selection**, Server Roles, Features, Confirmation, and Results. The main area is titled 'Select destination server' and contains the instruction: 'Select a server or a virtual hard disk on which to install roles and features.' There are two radio buttons: 'Select a server from the server pool' (selected) and 'Select a virtual hard disk'. Below this is a 'Server Pool' table with a filter box and a table with columns 'Name', 'IP Address', and 'Operating System'. The table contains one entry: 'WIN-SERVER', '192.168.1.2', and 'Microsoft Windows Ser'. Below the table, it says '1 Computer(s) found' and 'This page shows servers that are running Windows Server 2012 or a new and that have been added by using the Add Servers command in Server Manager. Newly-added servers from which data collection is still incomplete are not shown.' At the bottom are '< Previous' and 'Next >' buttons.

Right Screenshot: Select server roles

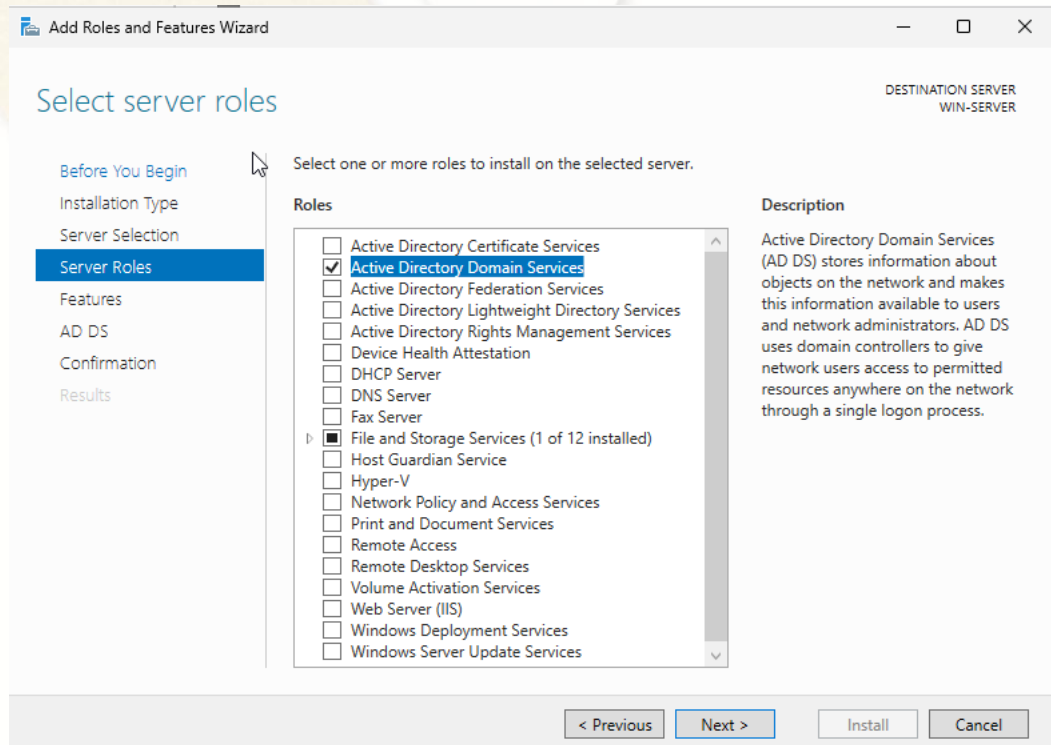
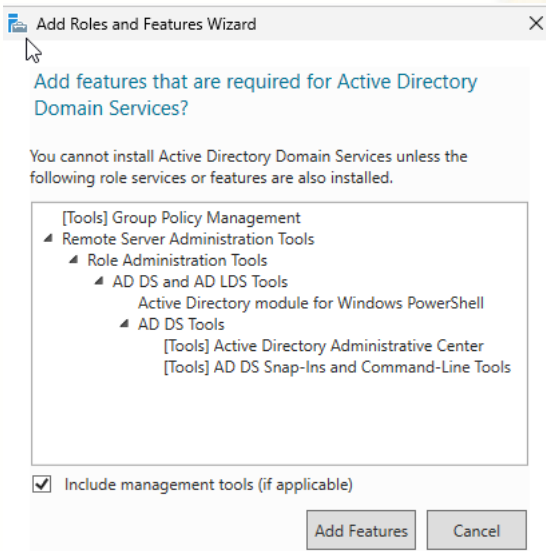
The wizard is titled 'Add Roles and Features Wizard'. The left sidebar shows the progression: Before You Begin, Installation Type, Server Selection, **Server Roles**, Features, Confirmation, and Results. The main area is titled 'Select server roles' and contains the instruction: 'Select one or more roles to install on the selected server.' There are two panes: 'Roles' and 'Description'. The 'Roles' pane has a list of roles with checkboxes. 'Active Directory Domain Services' is checked. The 'Description' pane shows the description for 'Active Directory Domain Services (AD DS)'. At the bottom are '< Previous', 'Next >', 'Install', and 'Cancel' buttons.

Name	IP Address	Operating System
WIN-SERVER	192.168.1.2	Microsoft Windows Ser

Roles	Description
☒ Active Directory Domain Services	Active Directory Domain Services (AD DS) stores information about objects on the network and makes this information available to users and network administrators. AD DS uses domain controllers to give network users access to permitted resources anywhere on the network through a single logon process.

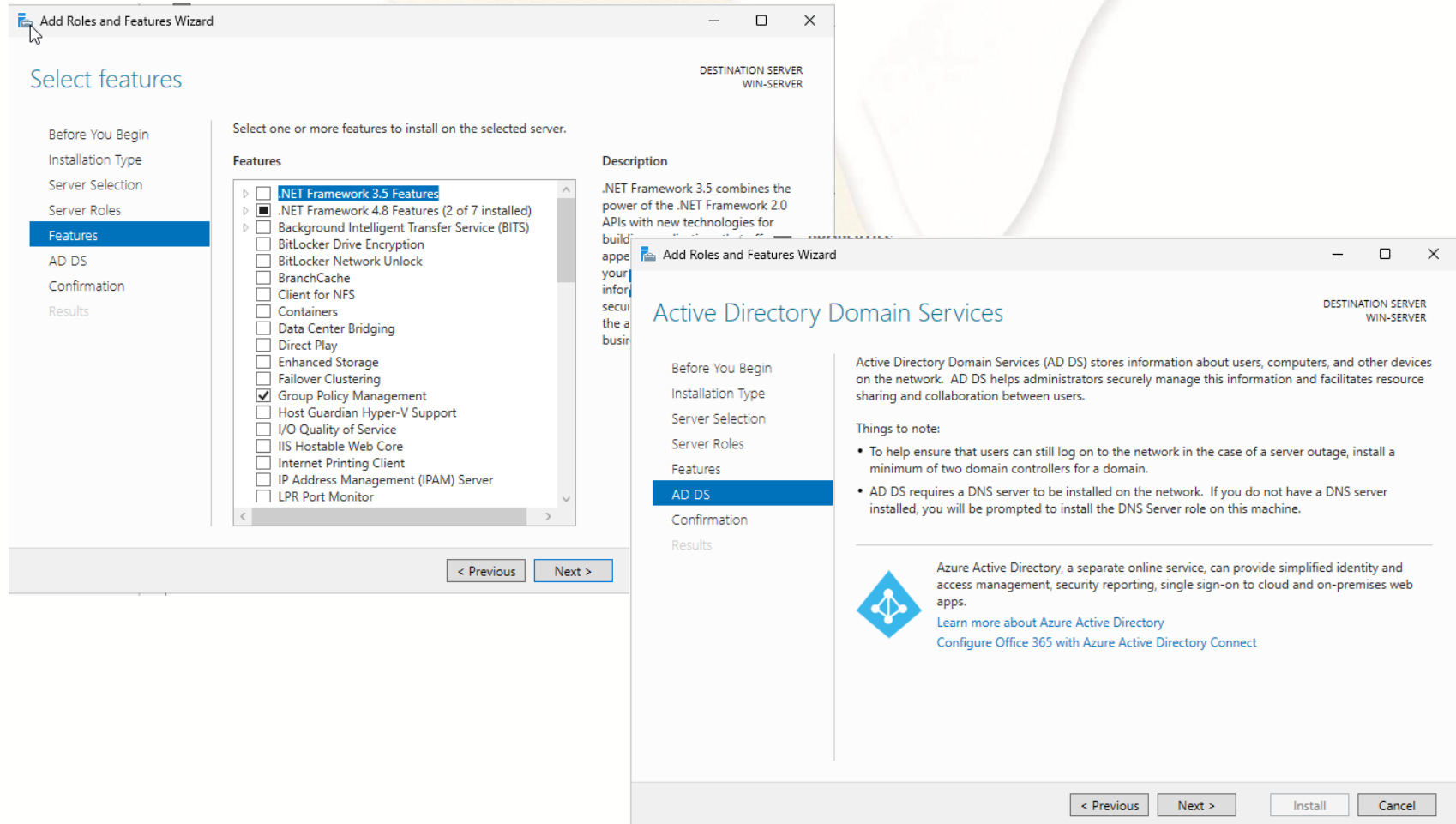
3.1. Installation (Cont.)

- Step 4: Click **Add Features** button > Click **Next**



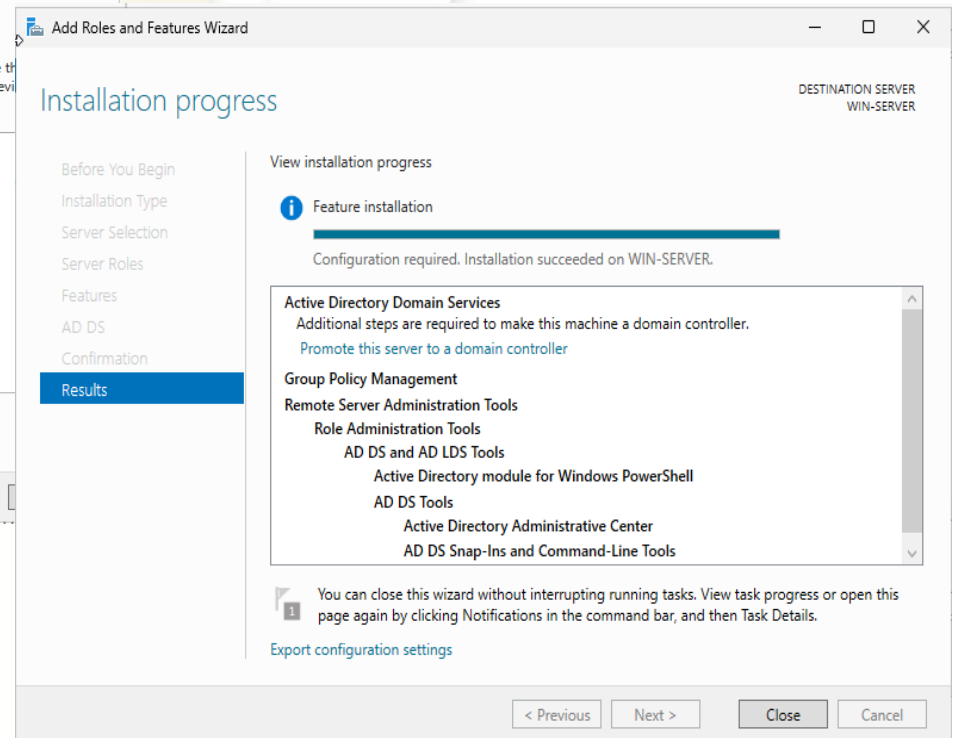
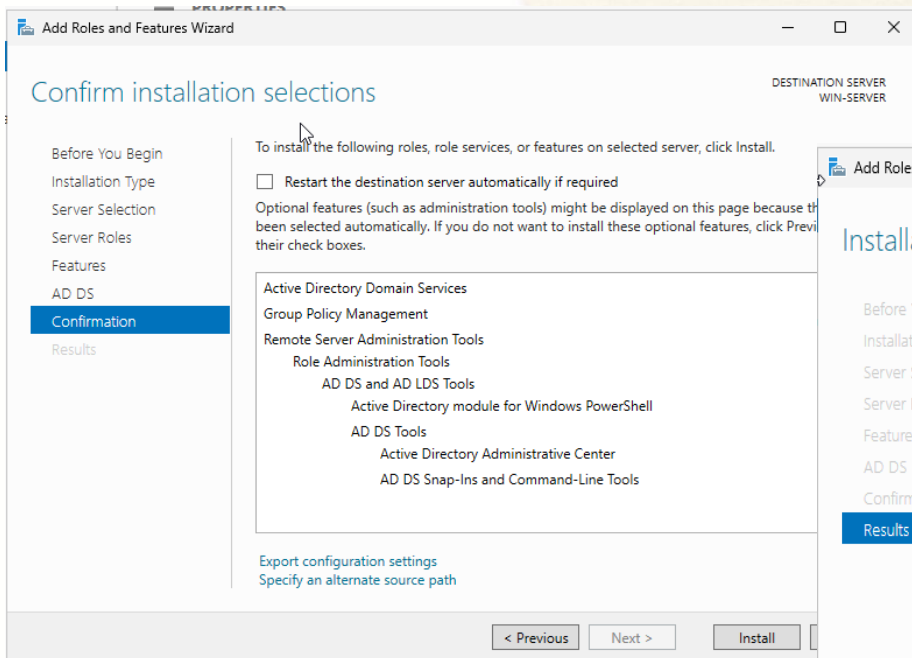
3.1. Installation (Cont.)

- Step 5: Click **Next** > Click **Next**



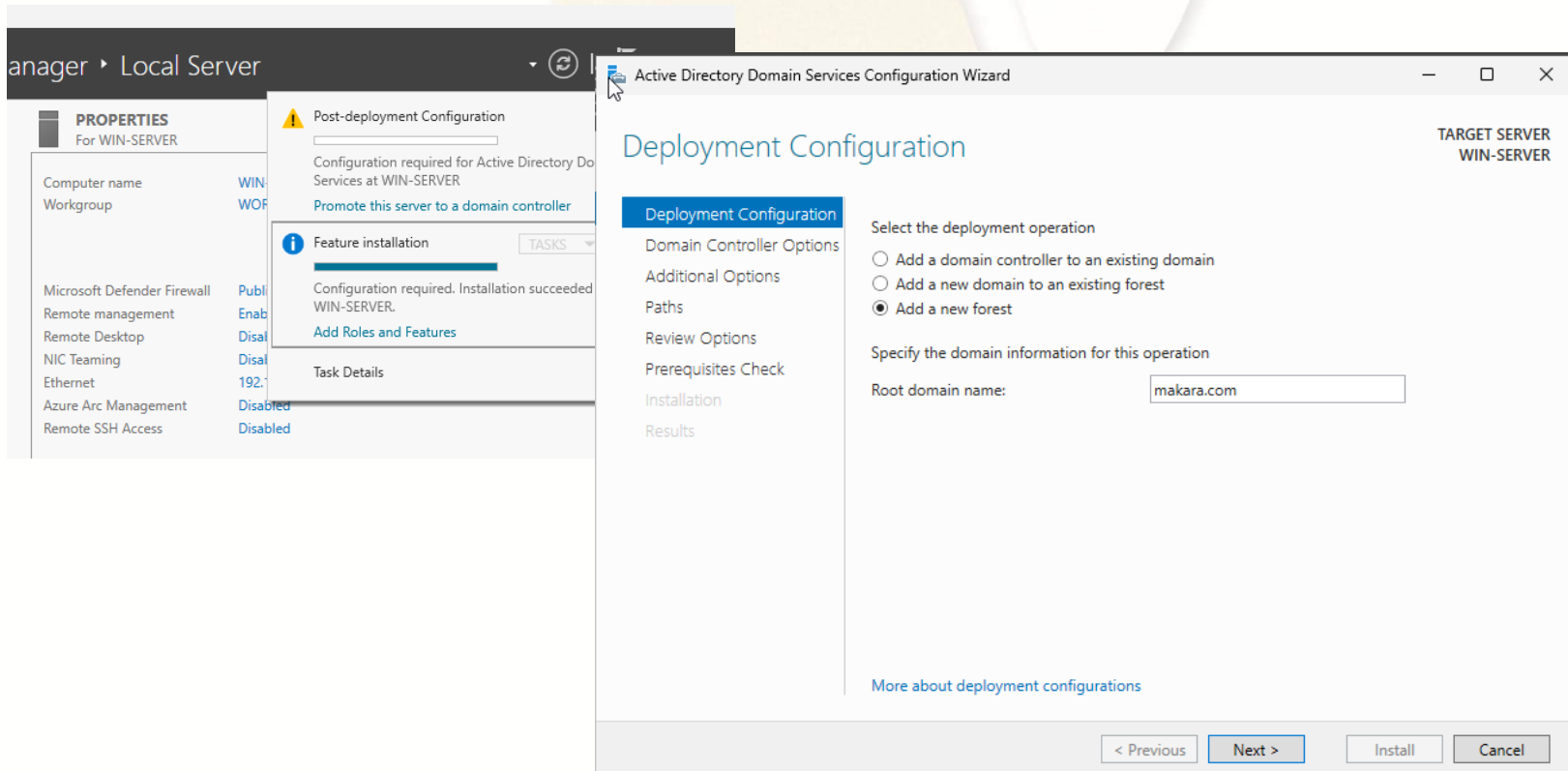
Installation (Cont.)

- Step 6: Click **Install** > *Wait until installation succeeded, Click **Close***



3.1. Installation (Cont.)

- Step 7: Click **Notification** menu > Click **Promote this server to a domain controller** link > Select **Add a new forest** > Enter Root domain name: (makara.com) > Click **Next**



3.1. Installation (Cont.)

- Step 8: Enter password twice in **Password** and **Confirm password** text fields (**for recovery process**) > Click **Next** > Click **Next**

The image displays two screenshots of the Active Directory Domain Services Configuration Wizard. The left screenshot shows the 'Domain Controller Options' step, where the user is prompted to select the functional level of the new forest and root domain (both set to 'Windows Server 2025'). It also allows specifying domain controller capabilities, with 'Domain Name System (DNS) server' and 'Global Catalog (GC)' checked. The user is then prompted to enter a password for the Directory Services Restore Mode (DSRM) and confirm it. The right screenshot shows the 'DNS Options' step, where the user is prompted to specify DNS delegation options. A warning message states: 'A delegation for this DNS server cannot be created because the authoritative parent zone cannot be found...'. The 'Create DNS delegation' checkbox is unchecked. Both screenshots show the 'Next >' button at the bottom.

Active Directory Domain Services Configuration Wizard

Domain Controller Options

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select functional level of the new forest and root domain

Forest functional level: Windows Server 2025

Domain functional level: Windows Server 2025

Specify domain controller capabilities

☒ Domain Name System (DNS) server

☒ Global Catalog (GC)

☐ Read only domain controller (RODC)

Type the Directory Services Restore Mode (DSRM) password

Password:

Confirm password:

[More about domain controller options](#)

< Previous Next >

Active Directory Domain Services Configuration Wizard

DNS Options

TARGET SERVER WIN-SERVER

A delegation for this DNS server cannot be created because the authoritative parent zone cannot be found... [Show more](#) X

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Specify DNS delegation options

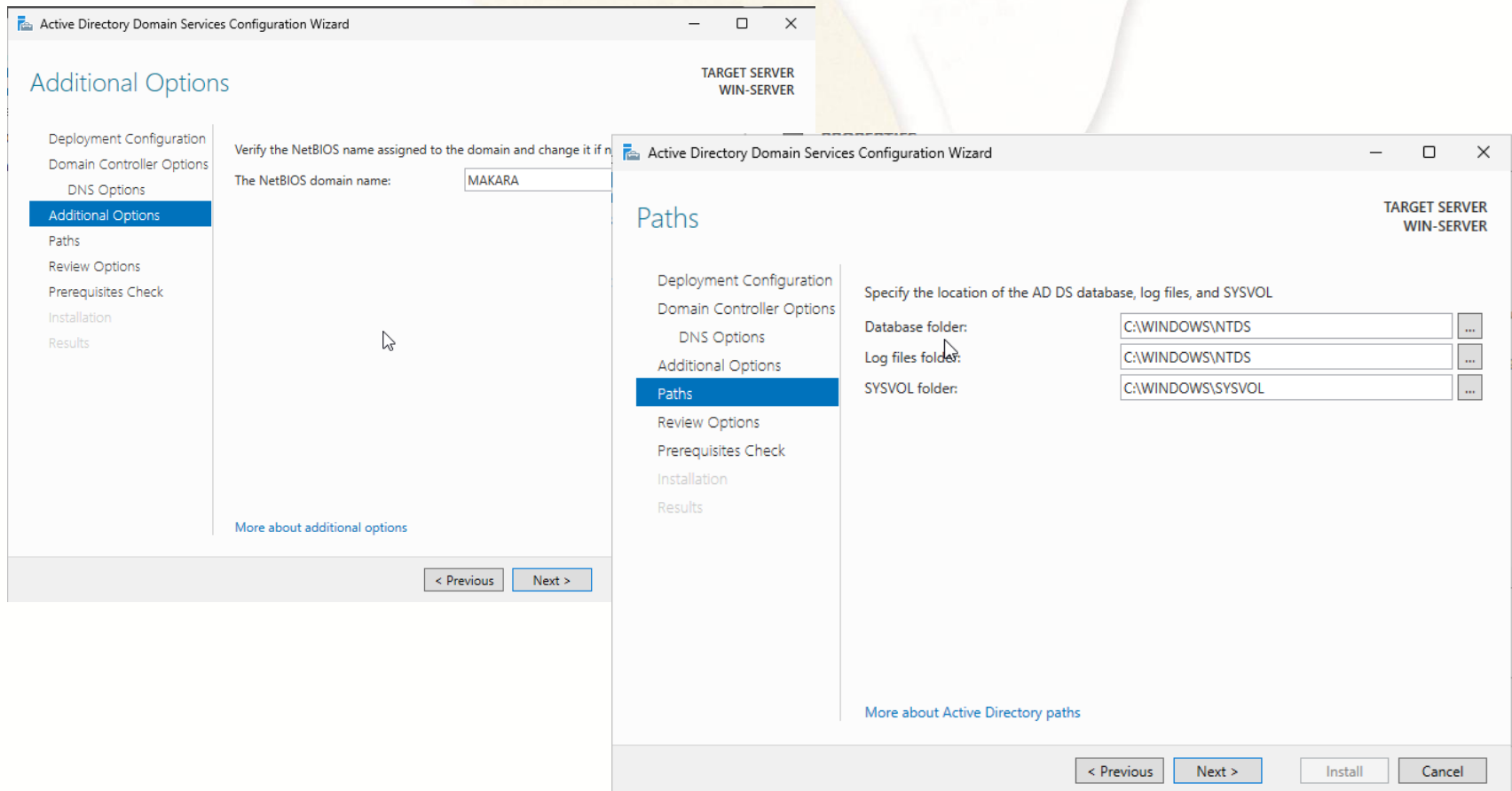
☐ Create DNS delegation

[More about DNS delegation](#)

< Previous Next > Install Cancel

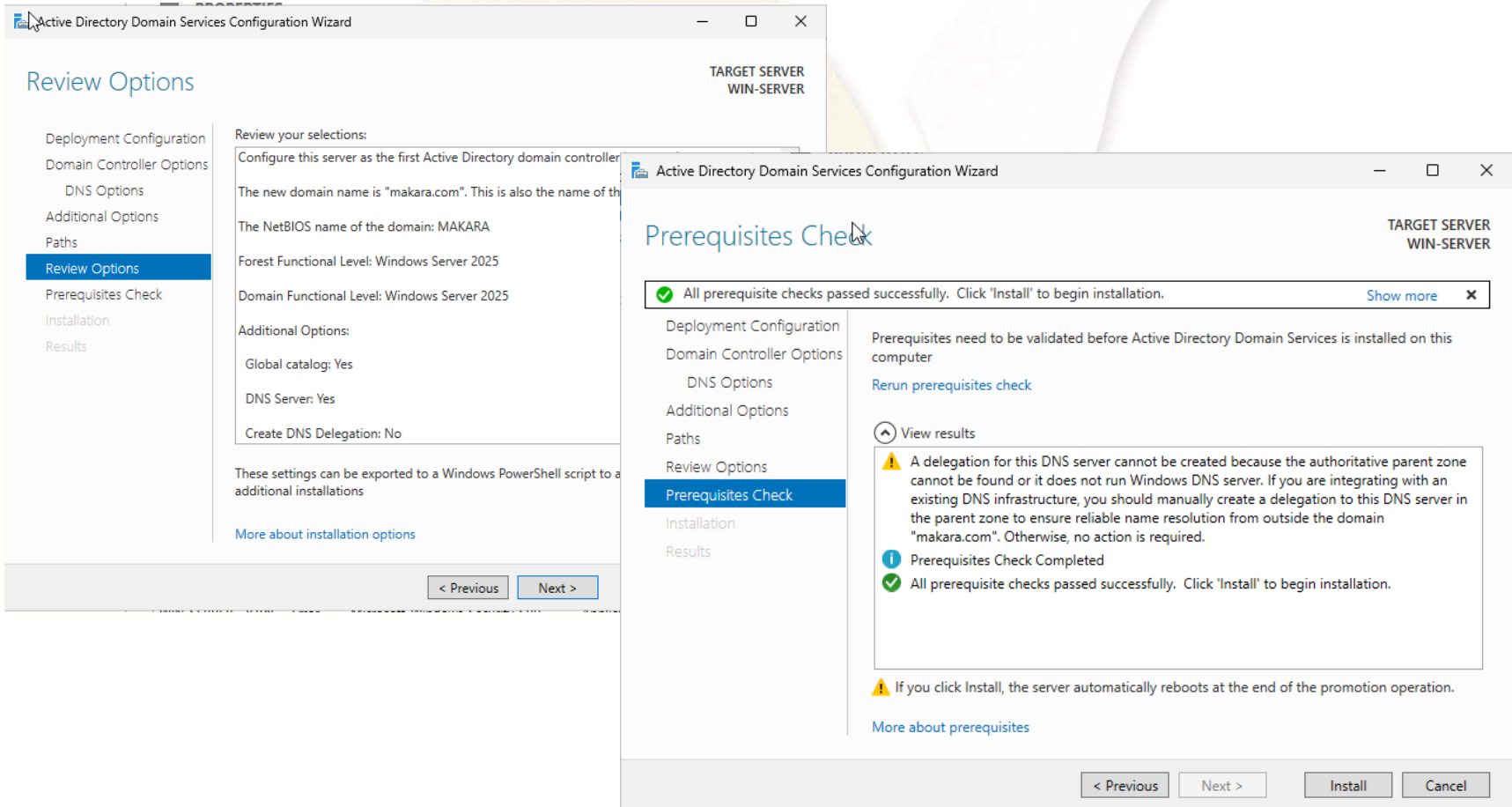
3.1. Installation (Cont.)

- Step 9: The **NetBIOS domain name** will be generated by itself, just click **Next** button > Click **Next**



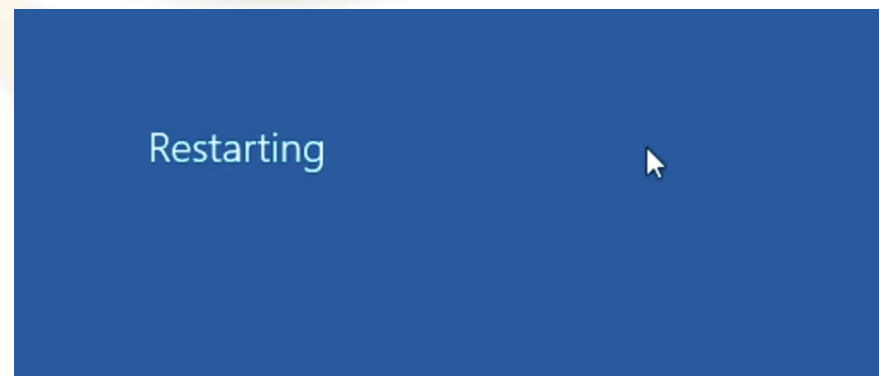
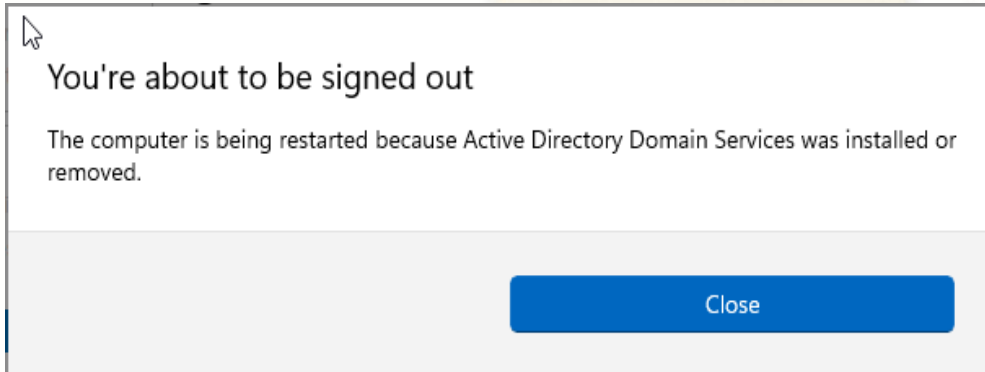
3.1. Installation (Cont.)

- Step 10: Click **Next** > Click **Install**



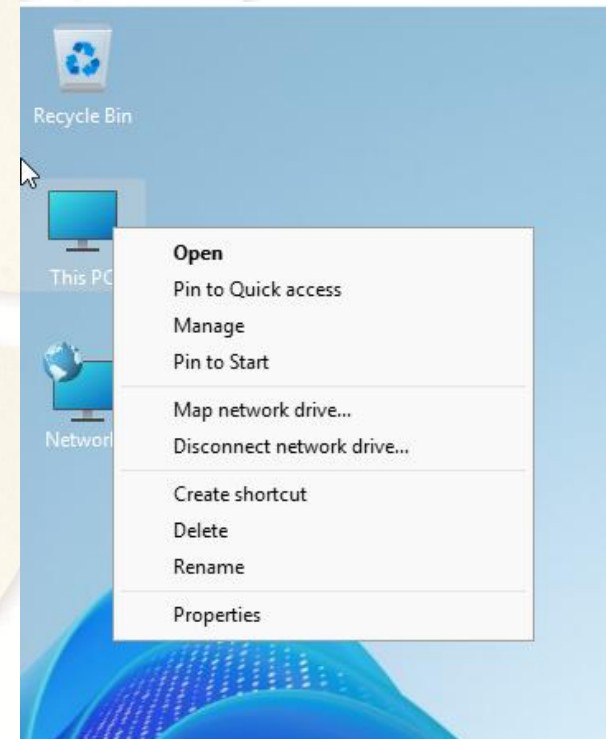
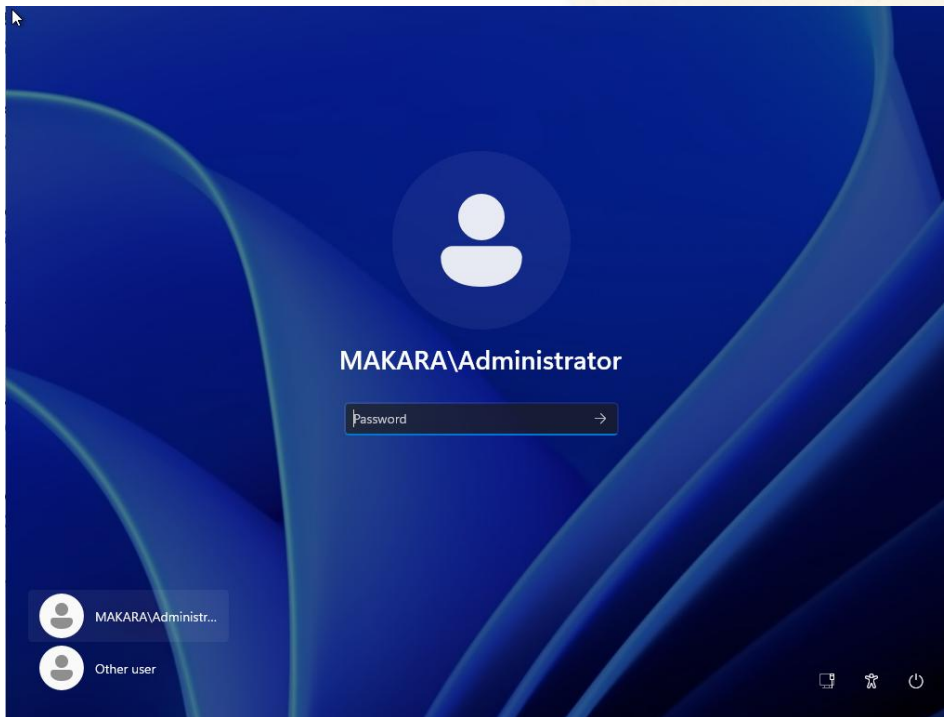
3.1. Installation (Cont.)

- Step 11: Server will **restart by itself**, just wait until it **starts up**



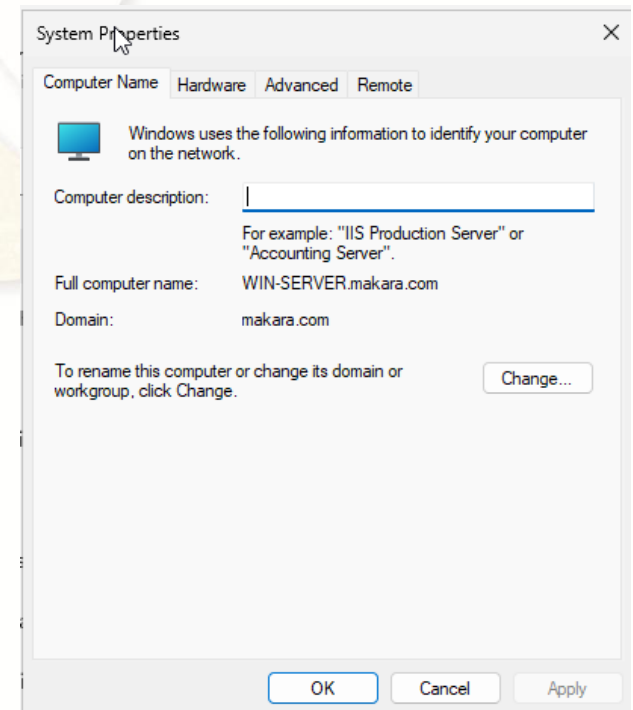
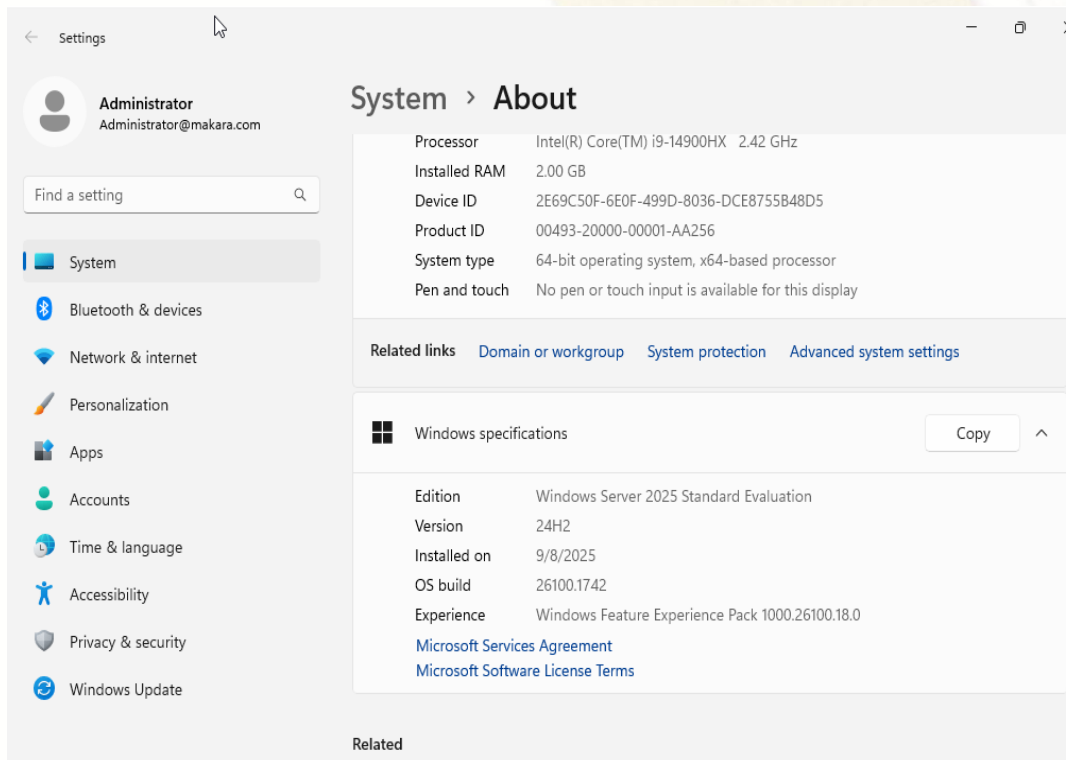
3.1. Installation (Cont.)

- Step 12: Enter username (MAKARA\Administrator) and password to login > Right click on This PC > Select Properties



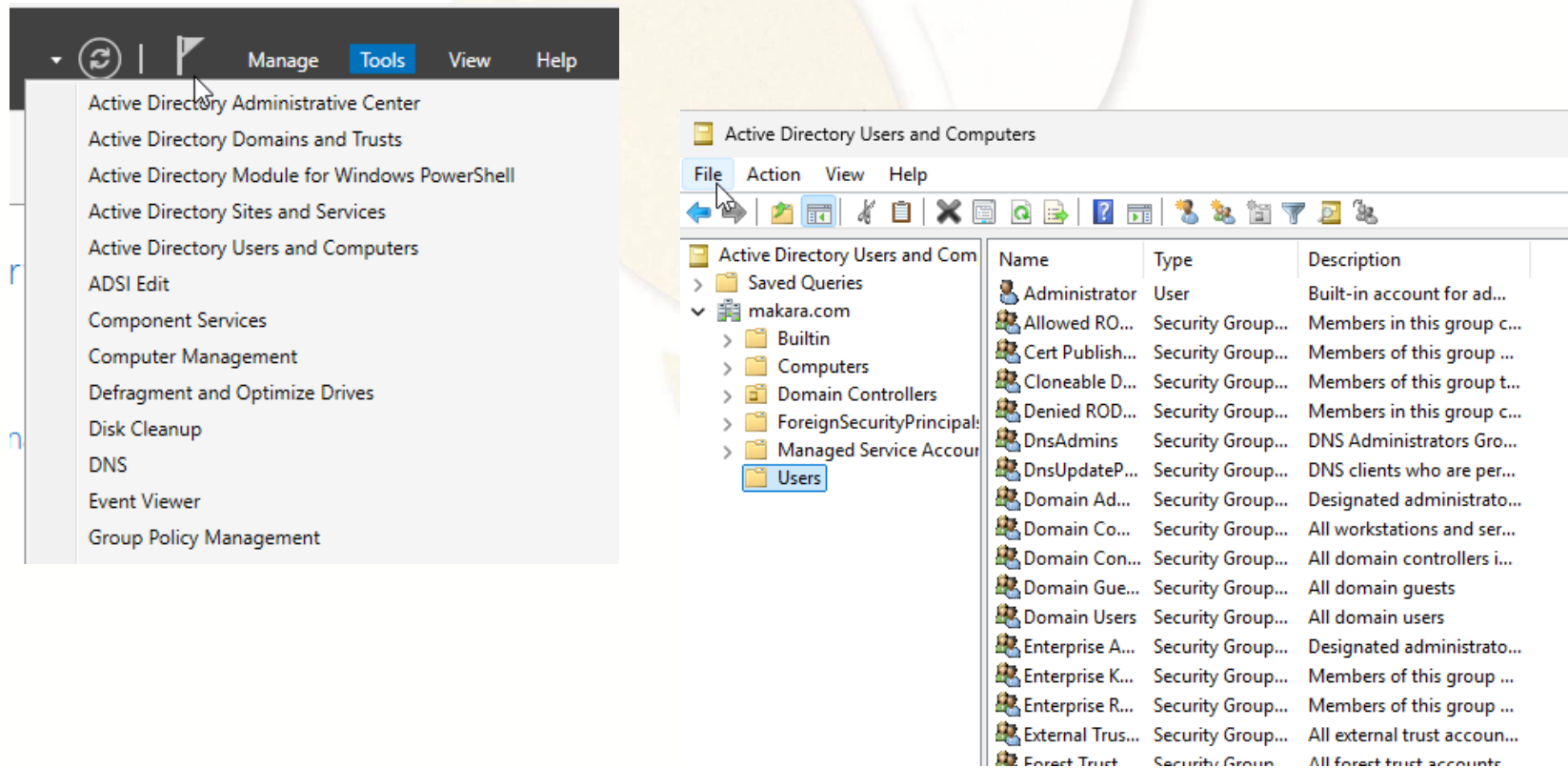
3.1. Installation (Cont.)

- Step 13: Click on **Domain or workgroup** you will see Full computer name: **WIN-SERVER.makara.com** and on **Domain: makara.com**



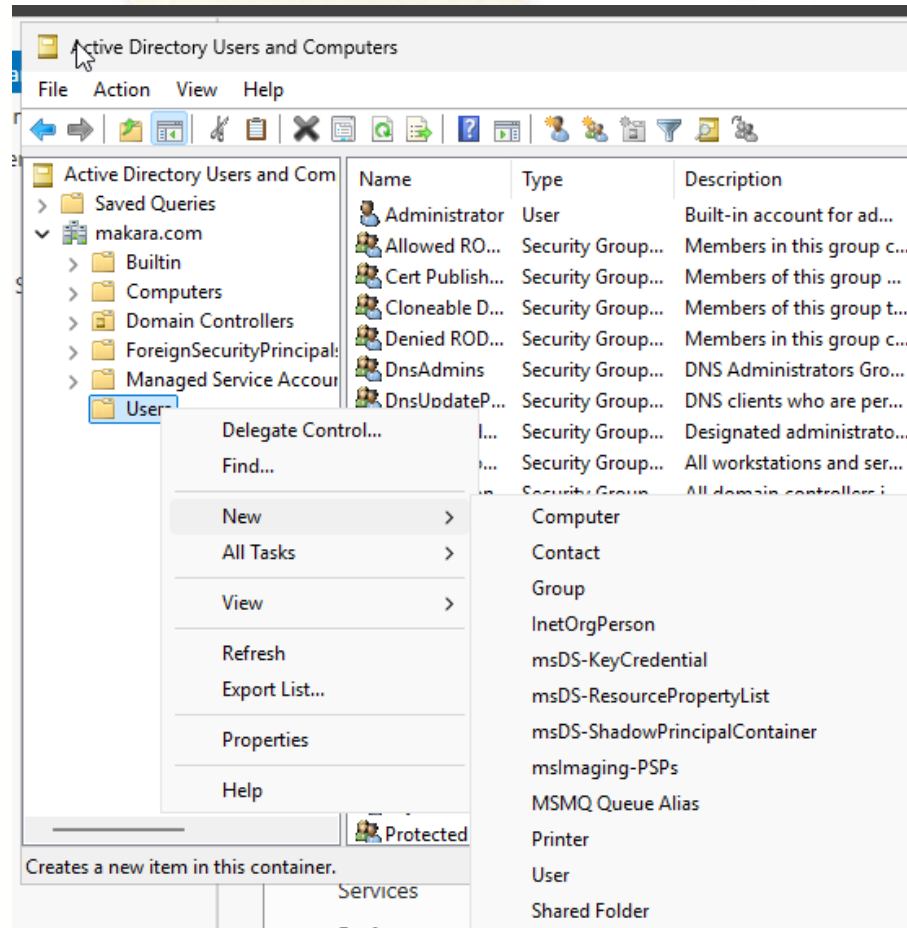
3.2. Create a User

- Step 1: Click **Tools** menu > Click **Active Directory Users and Computers** > Expand **domain name (makara.com)** > Select **Users** Folder



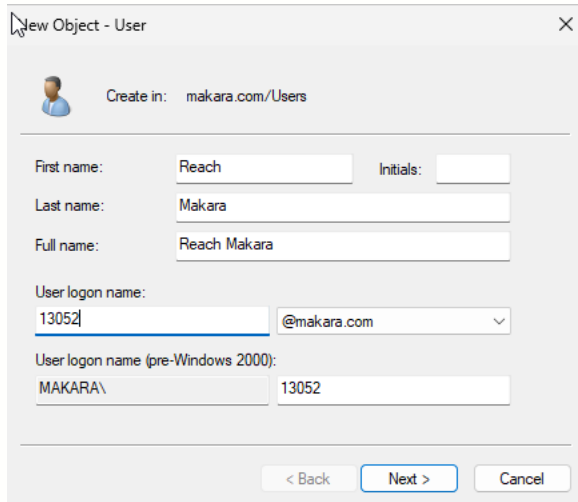
3.2. Create a User (Cont.)

- Step 2: Right click on user list > New > User



3.2. Create a User (Cont.)

- Step 3: Enter user information in **First name, Full name, User logon name** > Click **Next** > Enter password twice in **Password** and **Confirm password** > Click **Next** > Click **Finish**



New Object - User

Create in: makara.com/Users

First name: Reach Initials:

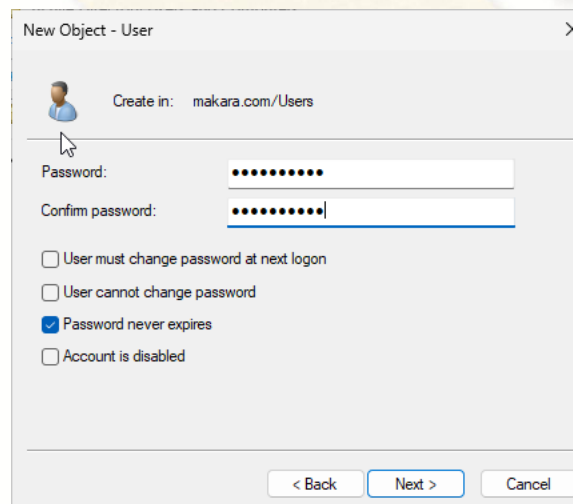
Last name: Makara

Full name: Reach Makara

User logon name: 13052 @makara.com

User logon name (pre-Windows 2000): MAKARA\ 13052

< Back Next > Cancel



New Object - User

Create in: makara.com/Users

Password:

Confirm password:

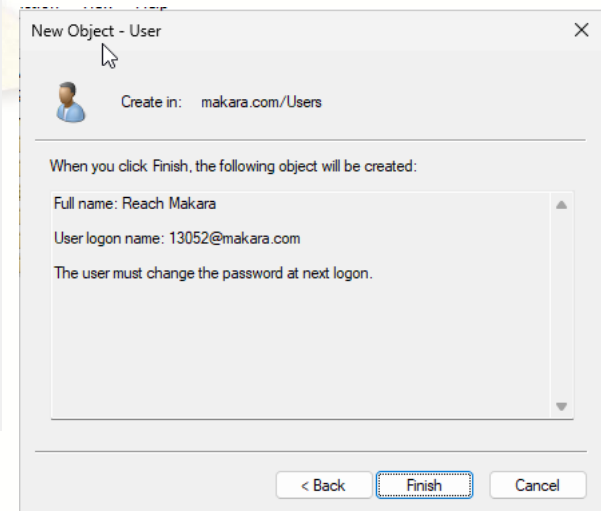
☐ User must change password at next logon

☐ User cannot change password

☒ Password never expires

☐ Account is disabled

< Back Next > Cancel



New Object - User

Create in: makara.com/Users

When you click Finish, the following object will be created:

Full name: Reach Makara

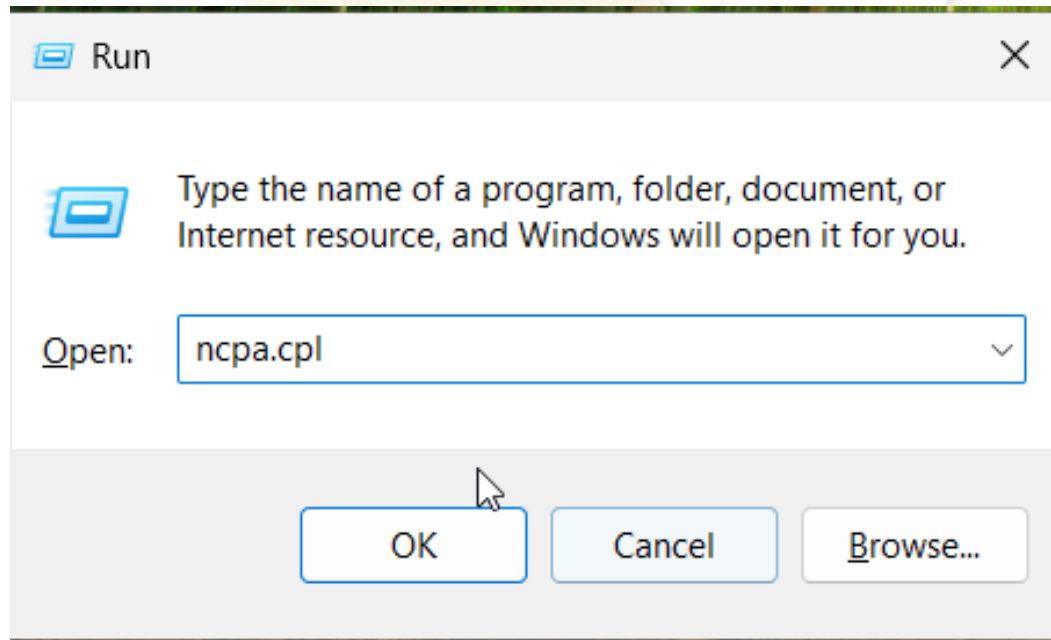
User logon name: 13052@makara.com

The user must change the password at next logon.

< Back Finish Cancel

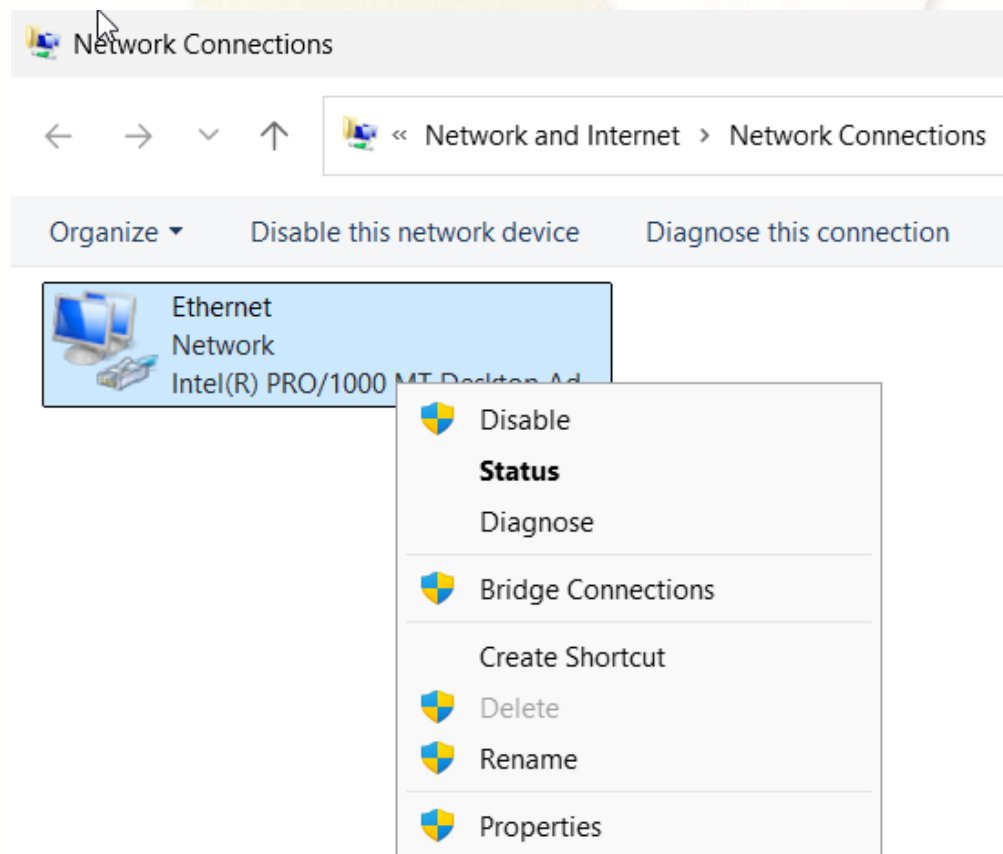
3.3. Join a Computer to a Domain

- Step 1: On **client computer**, go to **Run (Key Windows + R)** type **ncpa.cpl** > Click **OK** or hit **Enter**



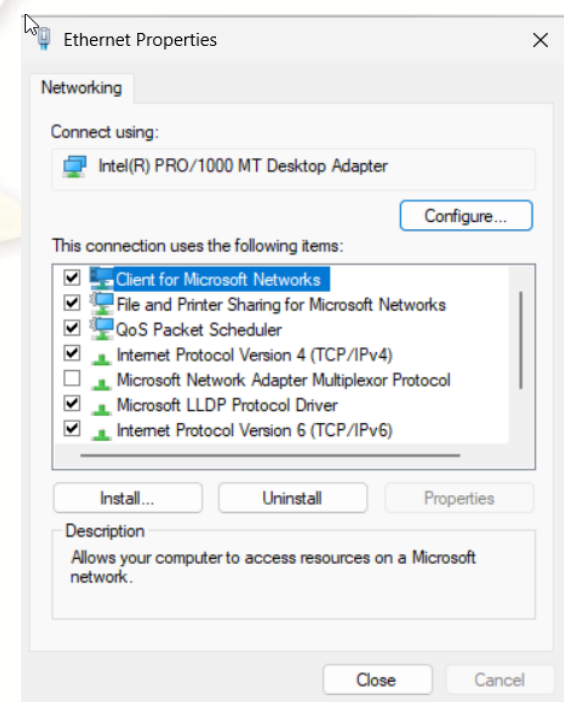
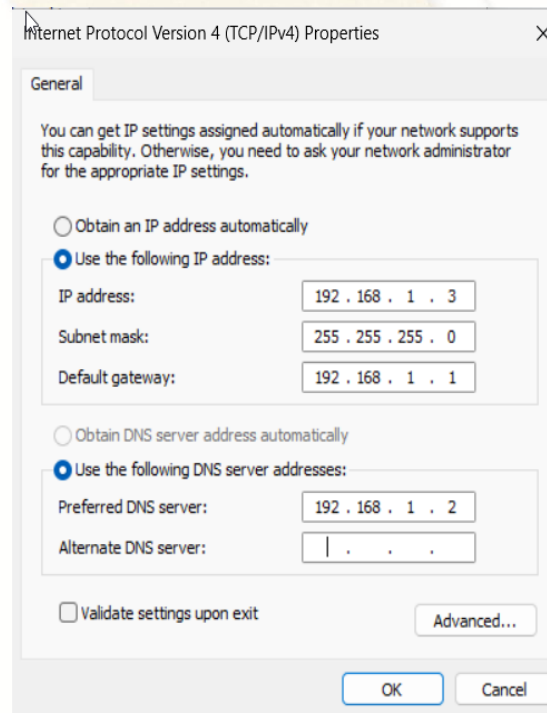
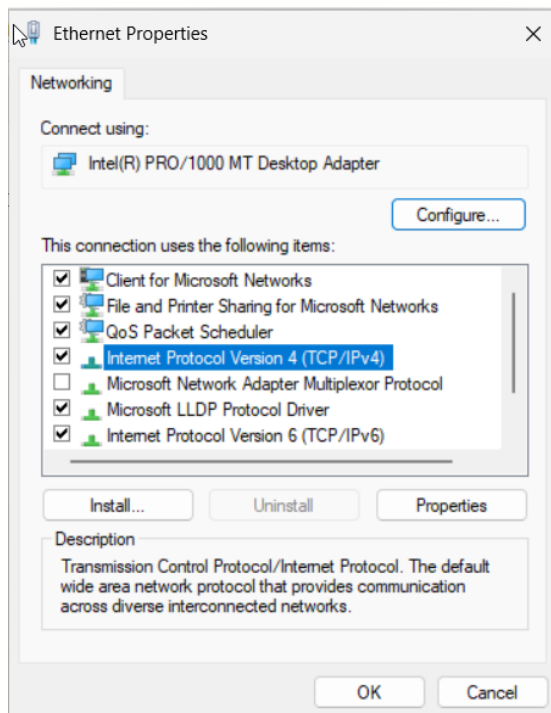
3.3. Join a Computer to a Domain (Cont.)

- Step 2: Right click on network adapter (Ethernet) > Select **Properties**



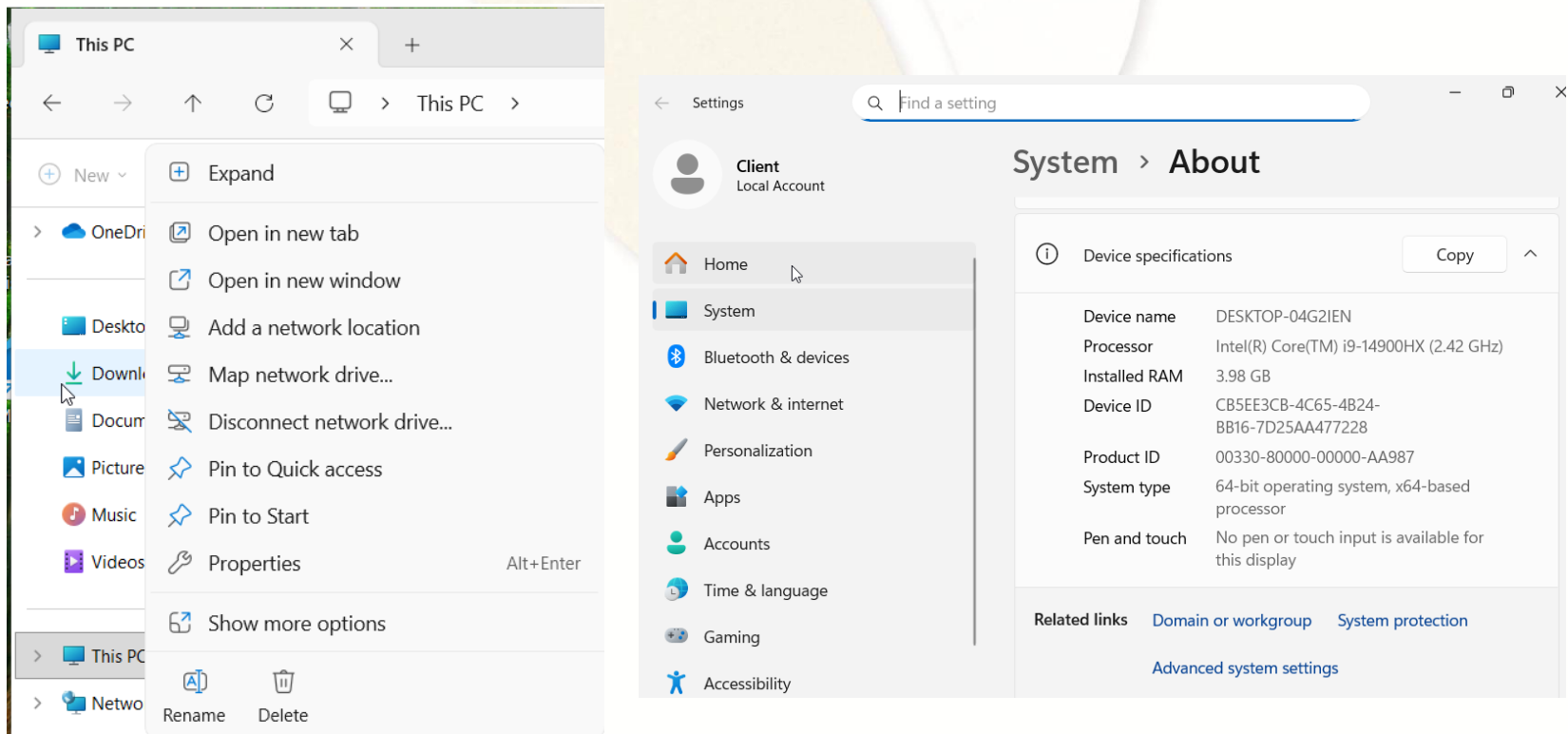
3.3. Join a Computer to a Domain (Cont.)

- Step 3: Select **Internet Protocol Version 4 (TCP/IPv4)** > Click **Properties** button > Enter **IP address**, **Subnet mask**, **Default gateway** and **Preferred DNS server** > Click **OK** > Click **Close**



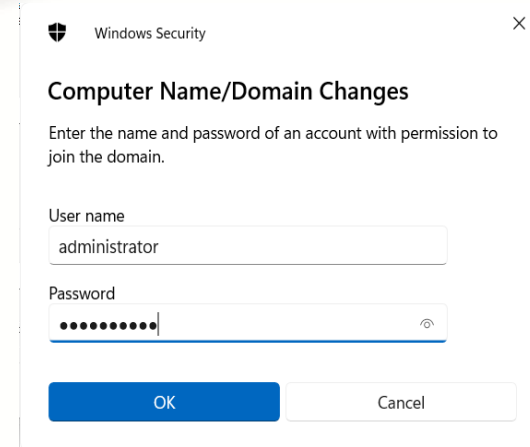
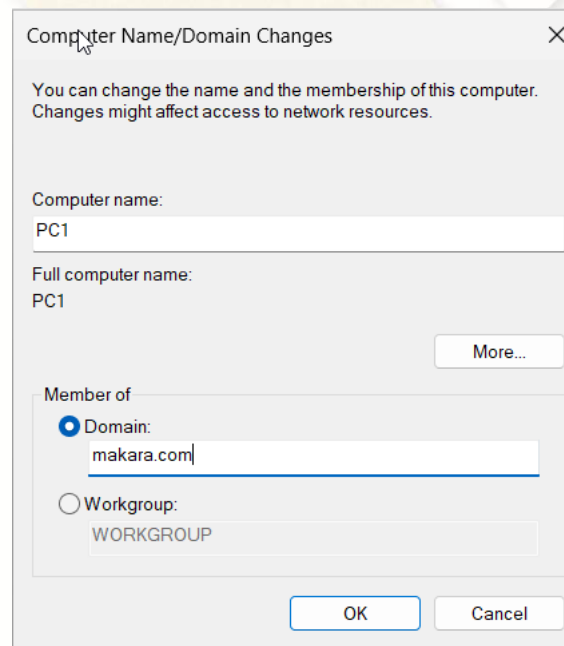
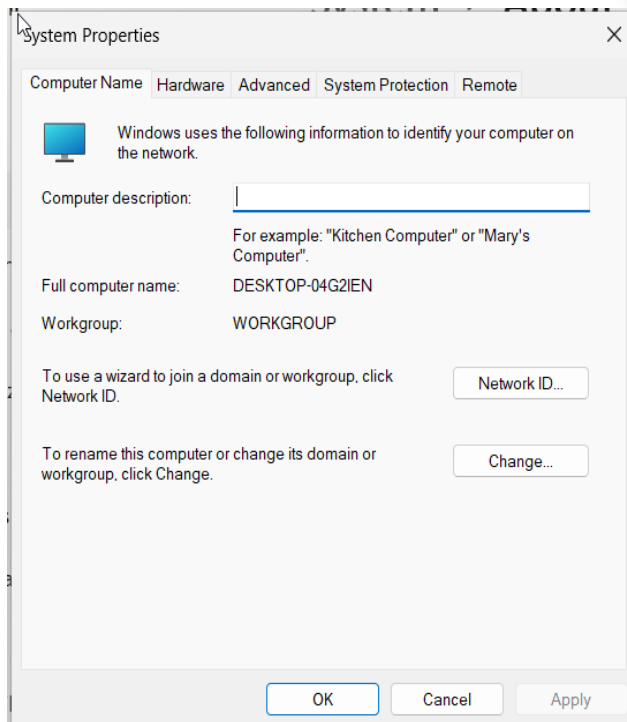
3.3. Join a Computer to a Domain (Cont.)

- Step 4: Right click on **This PC** > Select **Properties** > Click on **Domain or workgroup** link



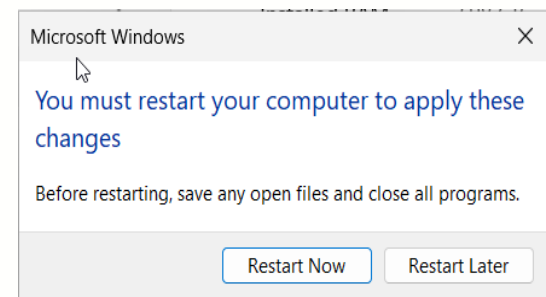
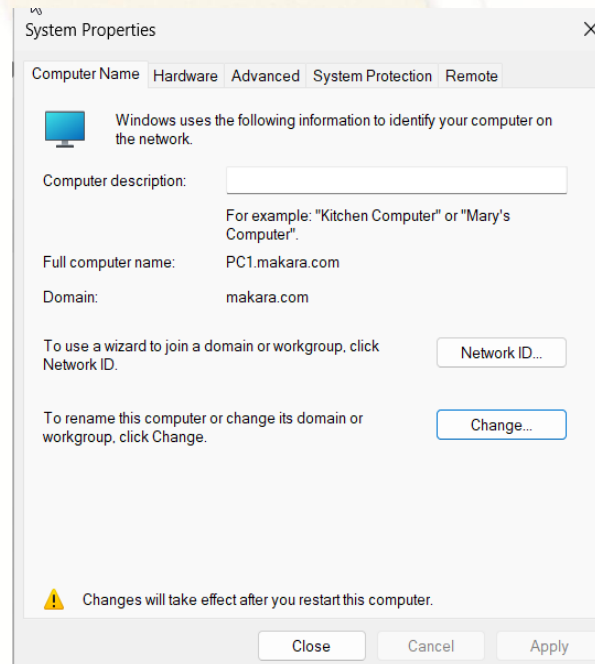
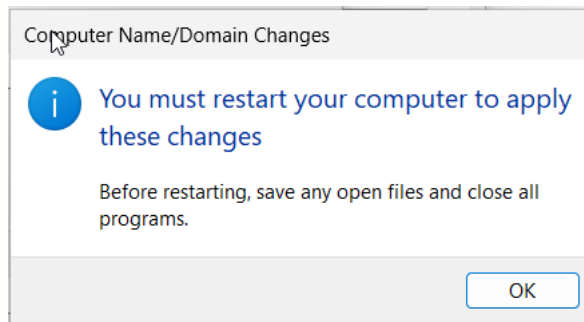
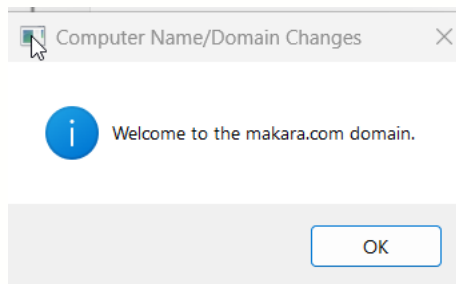
3.3. Join a Computer to a Domain (Cont.)

- Step 5: Click **Change** button > Select **Domain:** > Enter **domain name (makara.com)** > Click **OK** > Enter **Domain Administrator username and password** > Click **OK**



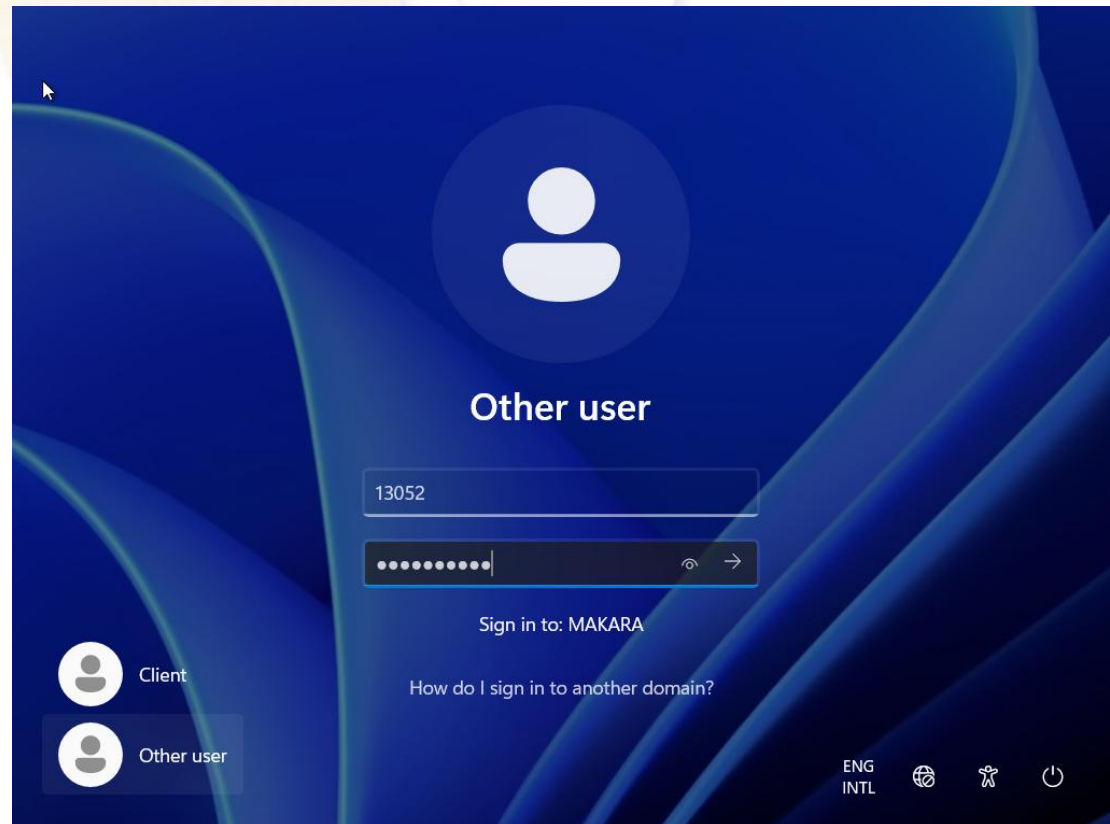
3.3. Join a Computer to a Domain (Cont.)

- Step 6: On **Computer Name/Domain Changes**, Click **OK** > Click **OK** > Click **Close** > Click **Restart Now** button



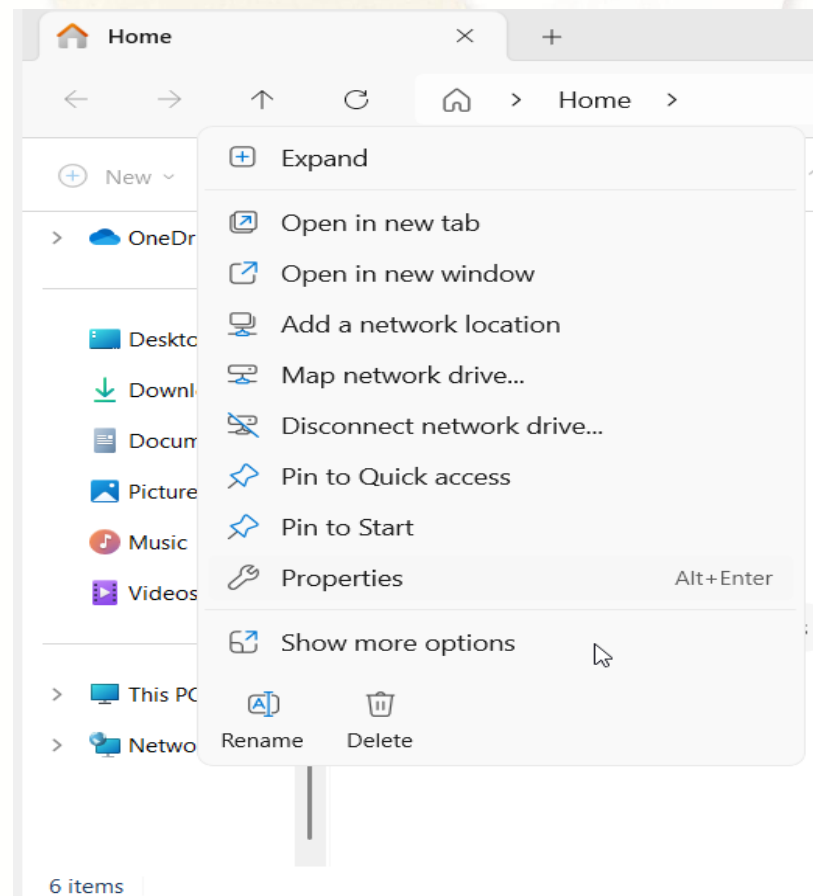
3.3. Join a Computer to a Domain (Cont.)

- Step 7: After **restart**, Click **Other user** > On **Username** please input **User** that create on Server (**Ex: 13052**) and **password** > Click **Submit** button or press **Enter** key



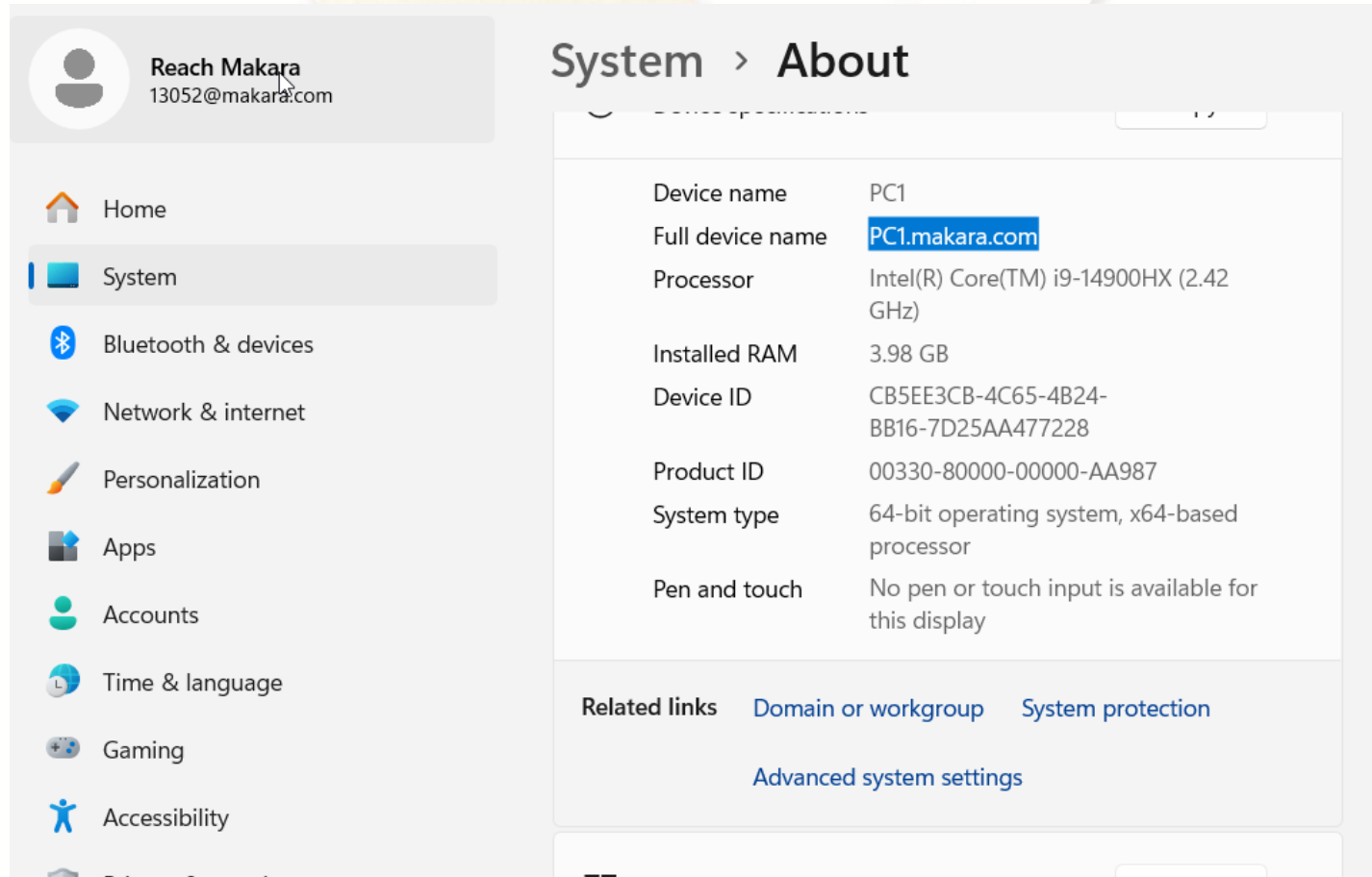
3.3. Join a Computer to a Domain (Cont.)

- Step 8: Press **Start + E key** to open **File Explorer** > Right click on **This PC** > Select **Properties**



3.3. Join a Computer to a Domain (Cont.)

- Step 9: You will see this computer name with domain

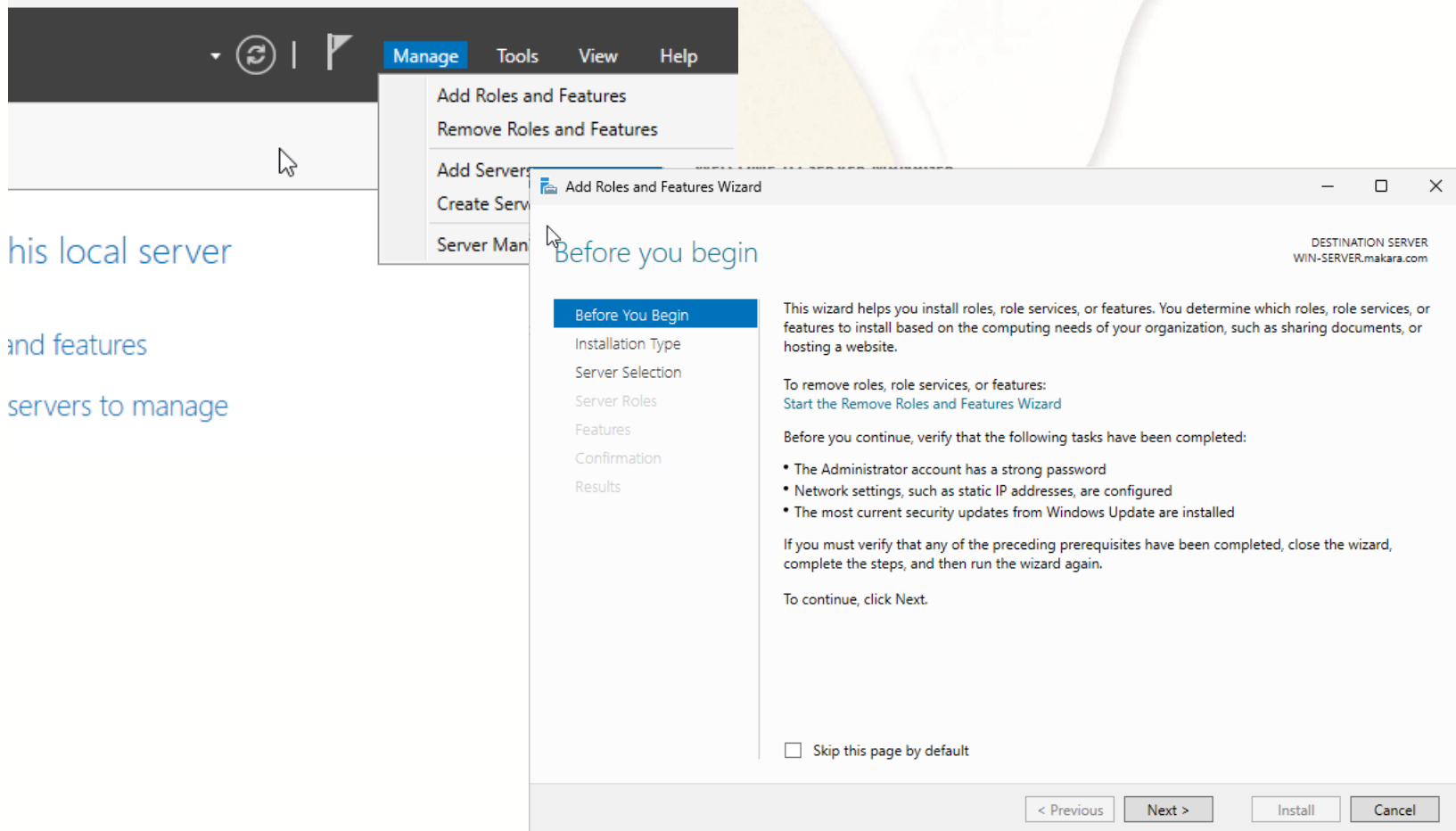




4. Backup Active Directory

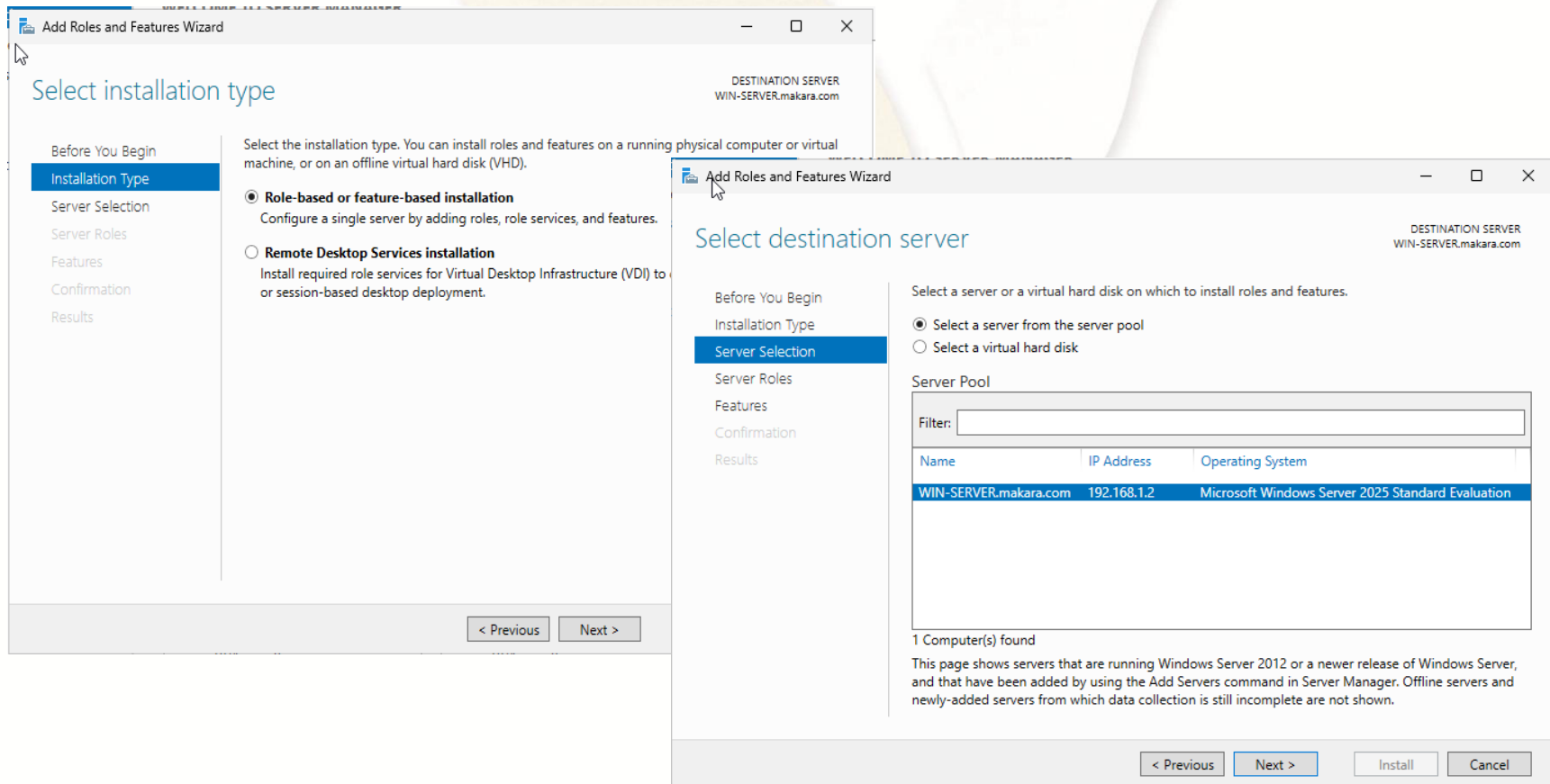
4.1. Install Windows Backup Feature

- Step 1: Click **Manage** menu > Click **Add Roles and Features** > Click **Next**



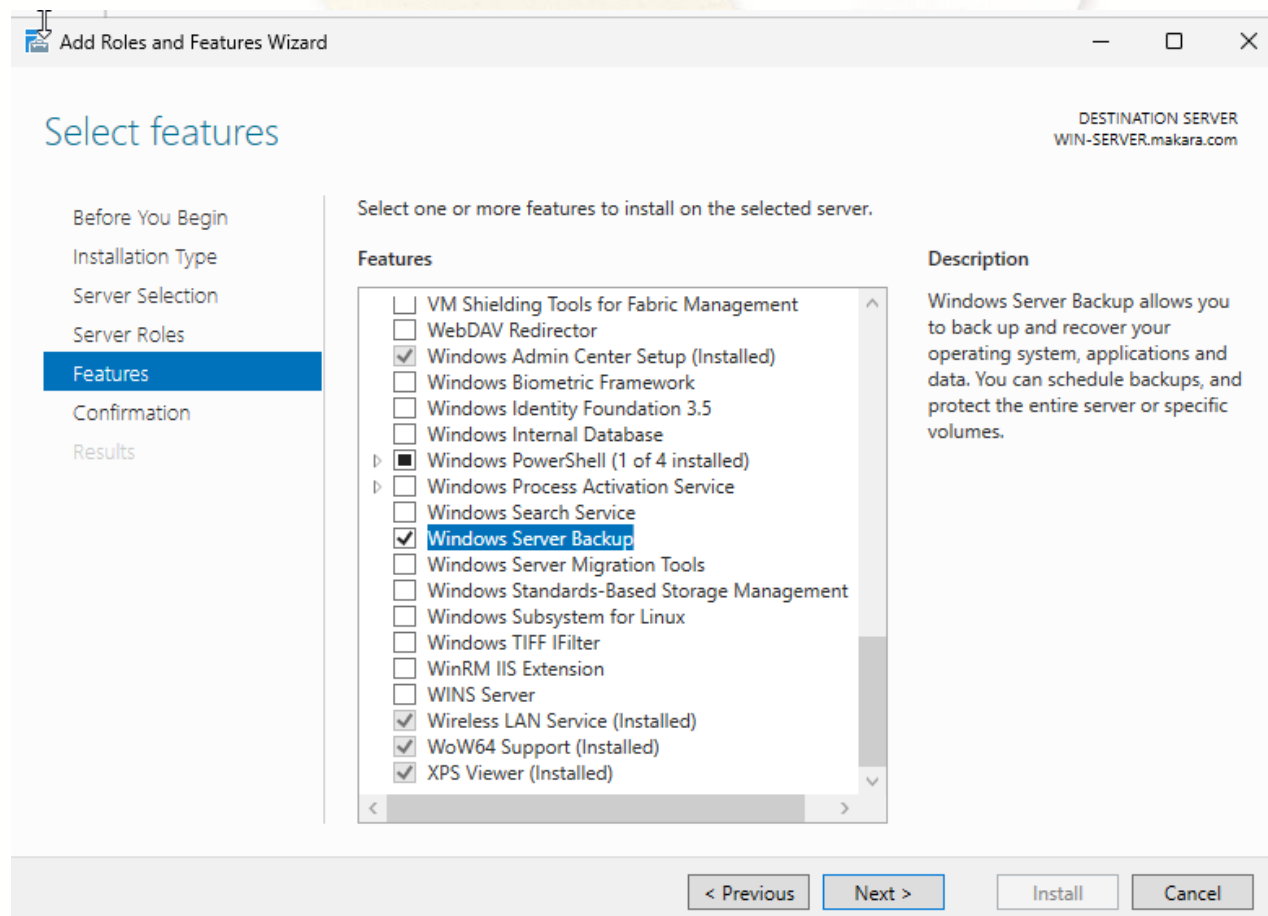
4.1. Install Windows Backup Feature (Cont.)

- Step 2: Click **Next** on **Select installation type** window > Click **Next**



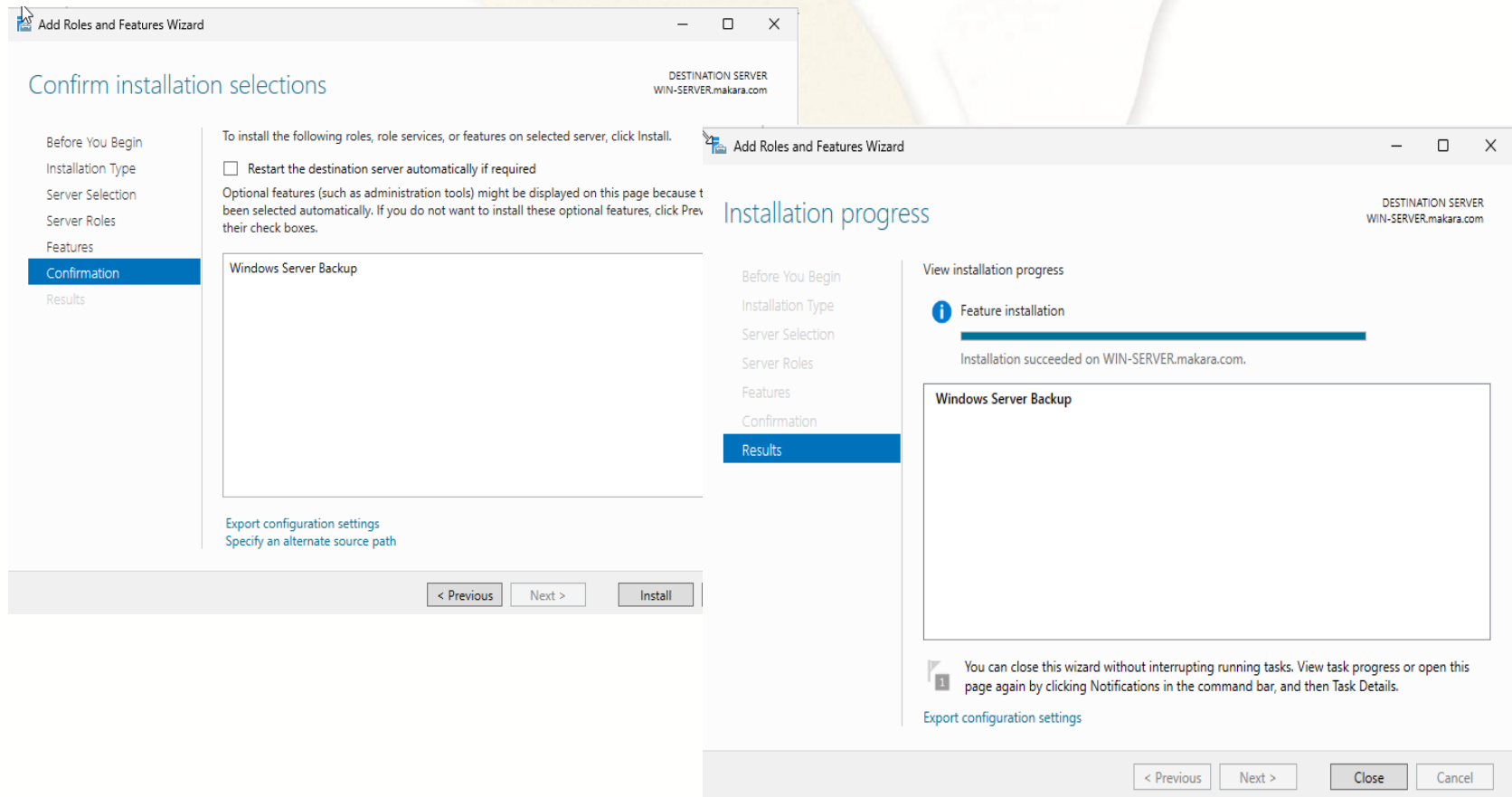
4.1. Install Windows Backup Feature (Cont.)

- Step 3: Select **Features** > Check **Windows Server Backup** > Click **Next**



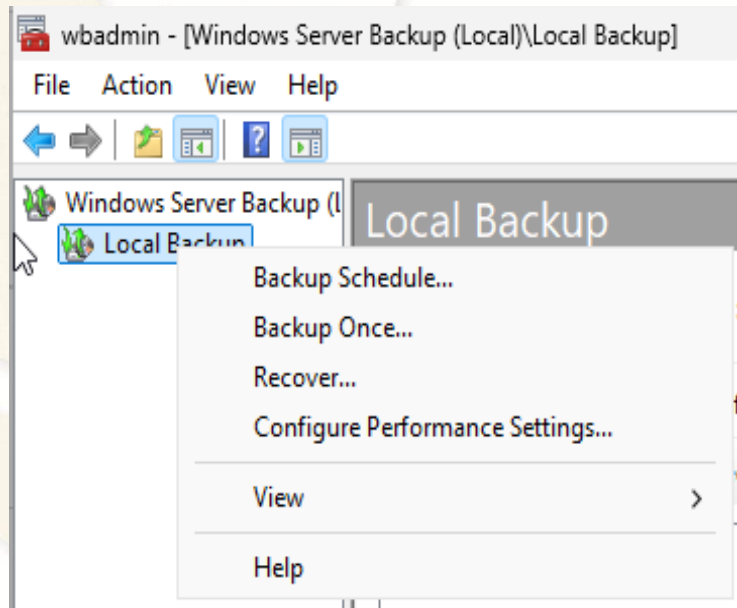
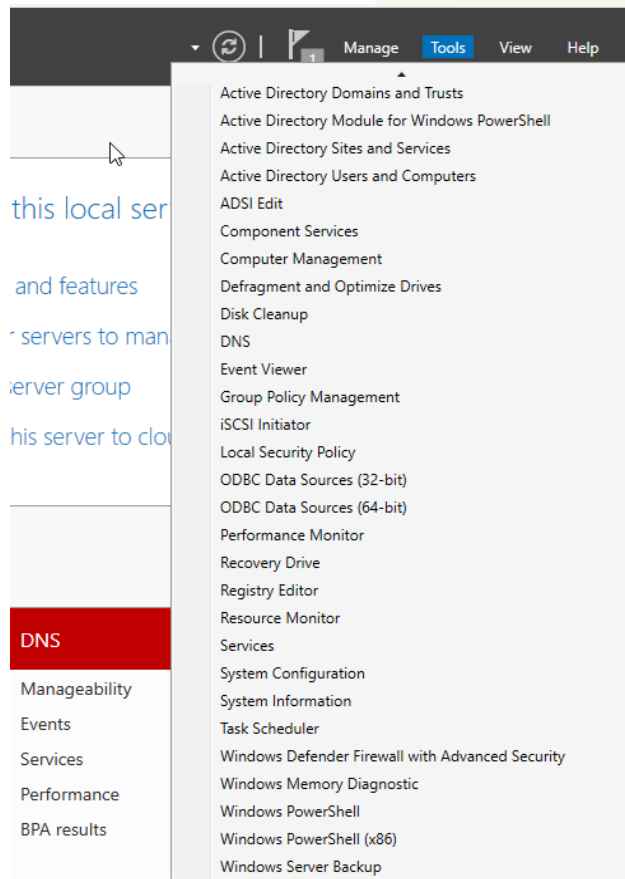
4.1. Install Windows Backup Feature (Cont.)

- Step 4: Click **Install** > wait **until installation succeeded**, click **Close** button



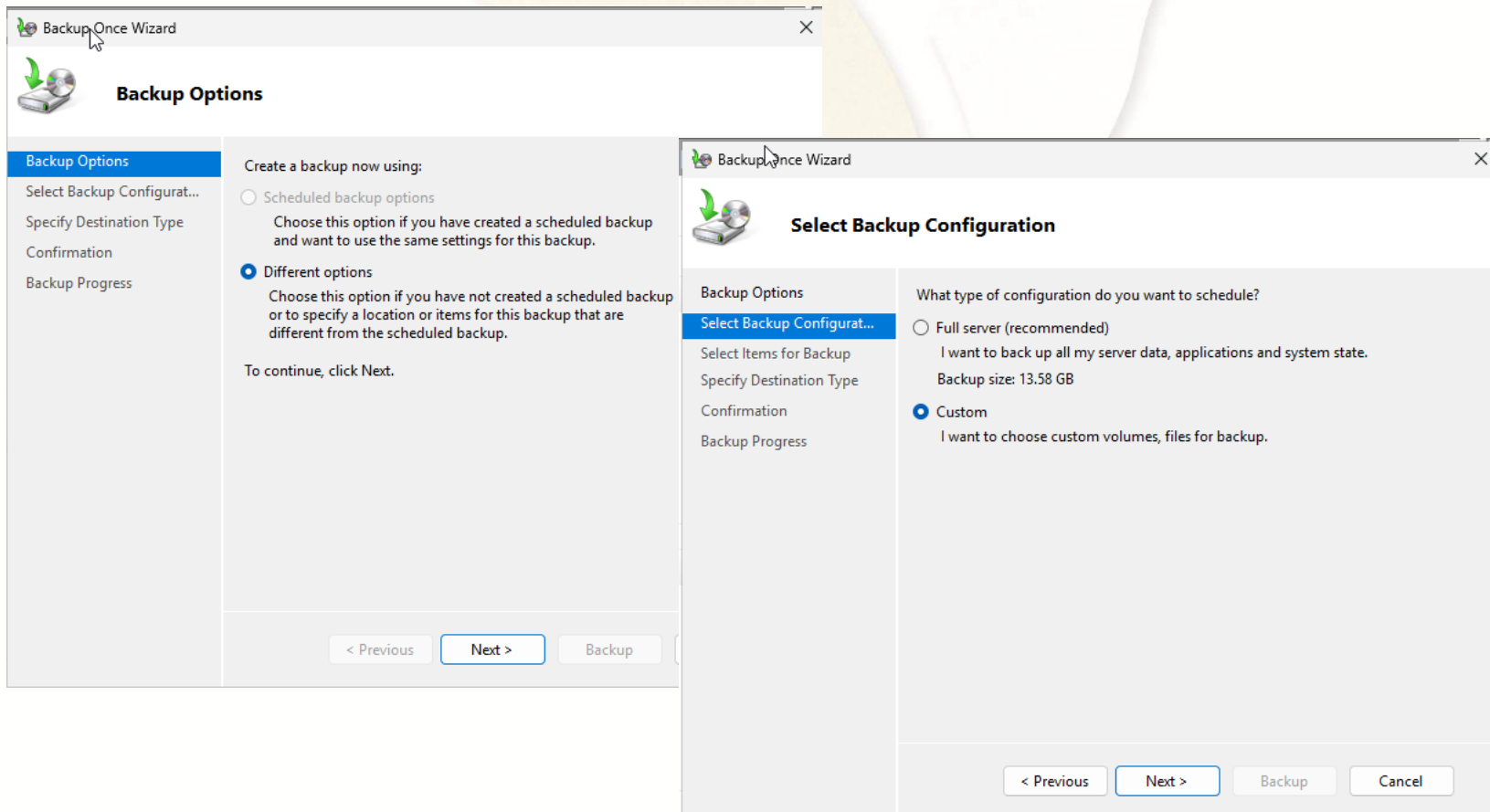
Backup Once AD

- Step 1: Click **Tools** menu and choose **Windows Server Backup** > Right click on **Local Backup** > Select **Backup Once...**



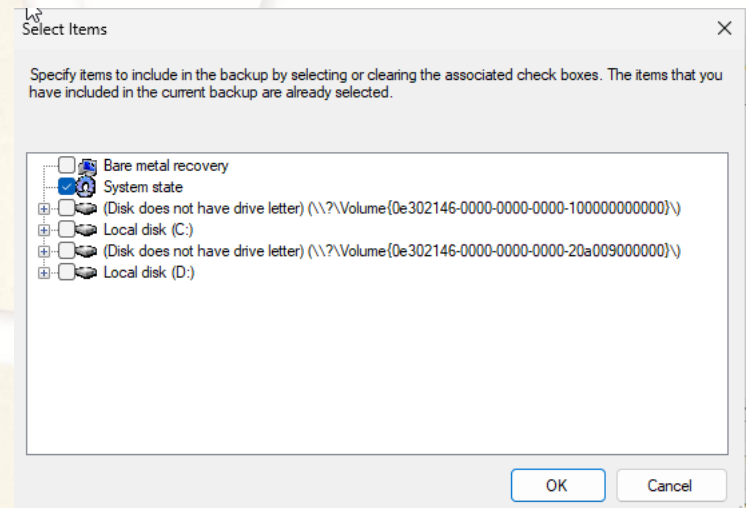
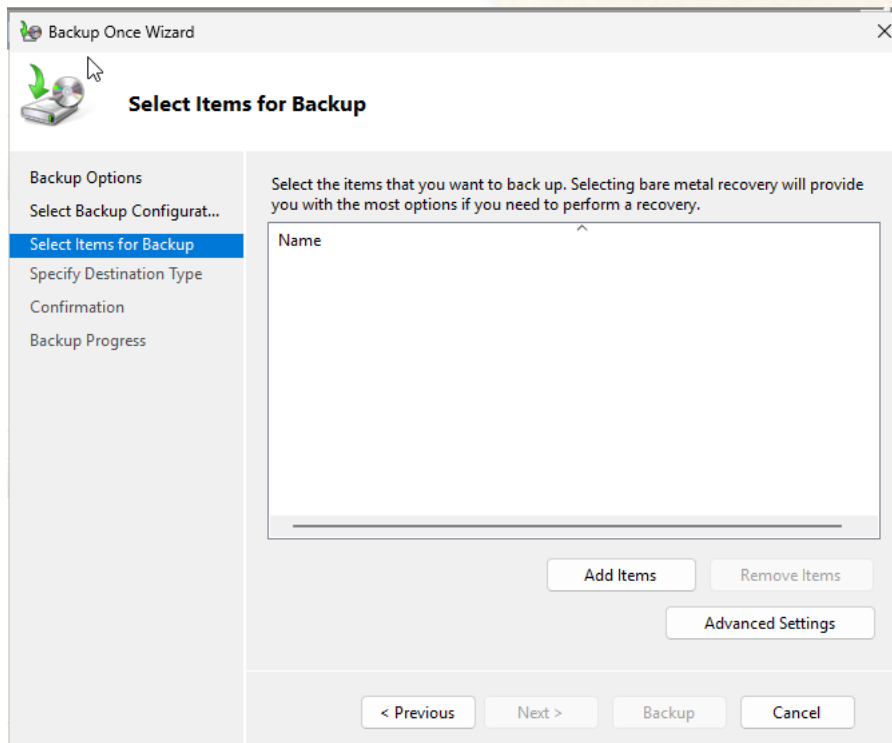
4.2. Backup Once AD (Cont.)

- Step 2: Select **Different options** > Click **Next** > Select **Custom** > Click **Next**



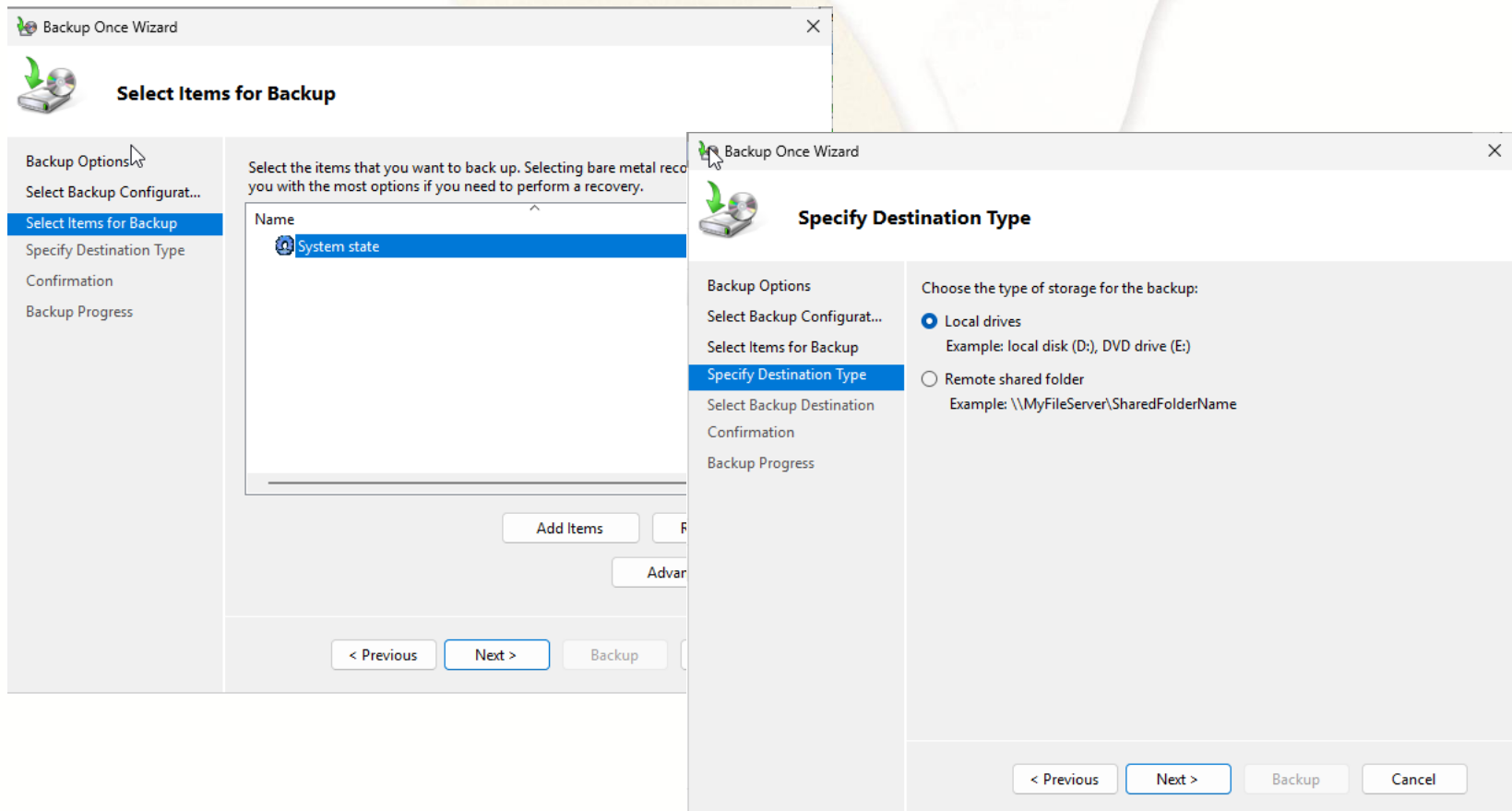
4.2. Backup Once AD (Cont.)

- Step 3: Click **Add Items** button > Choose **System state** > Click **OK**



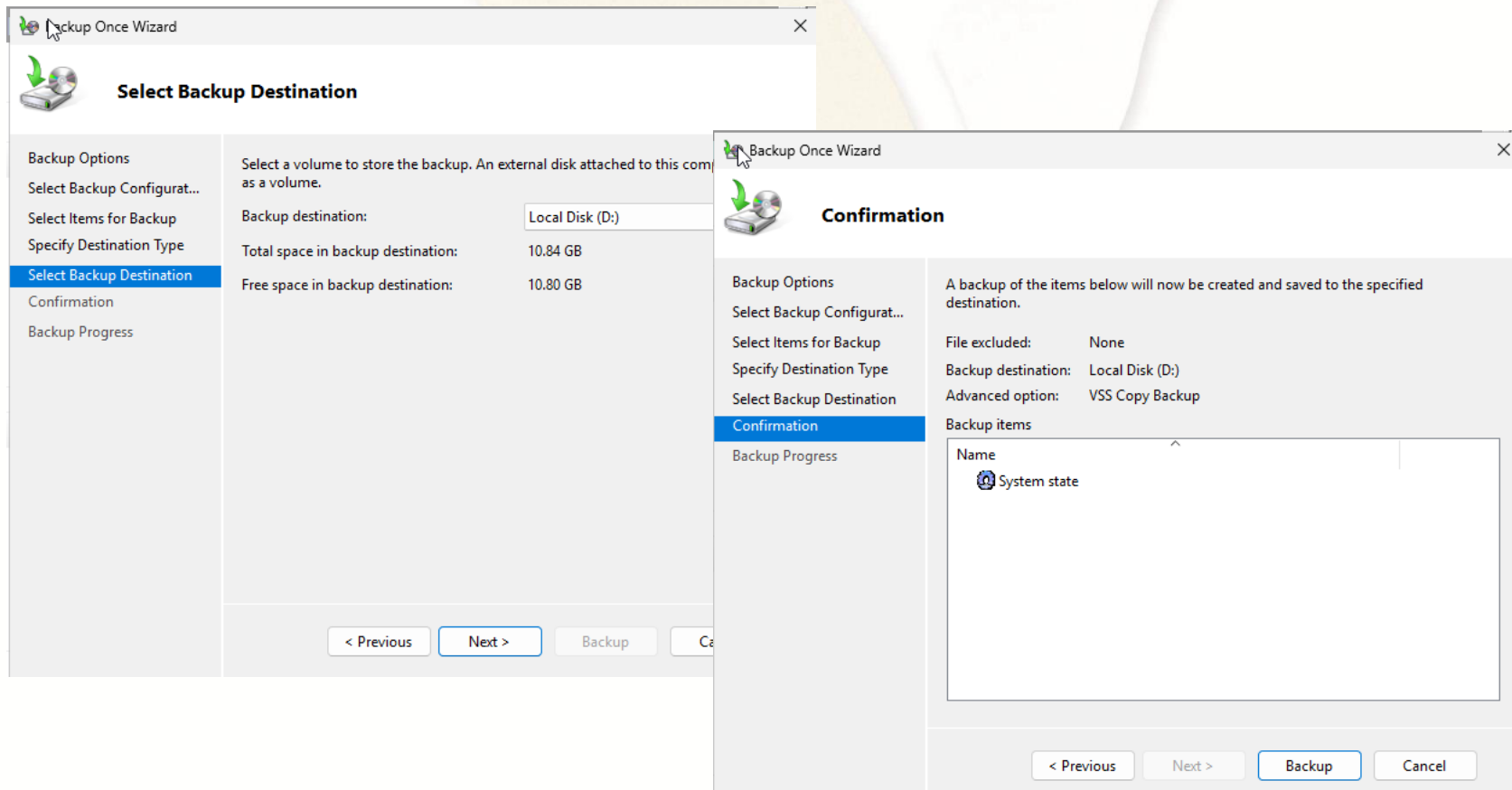
Backup Once AD (Cont.)

- Step 4: Click **Next** > Select **Local drives** > Click **Next**



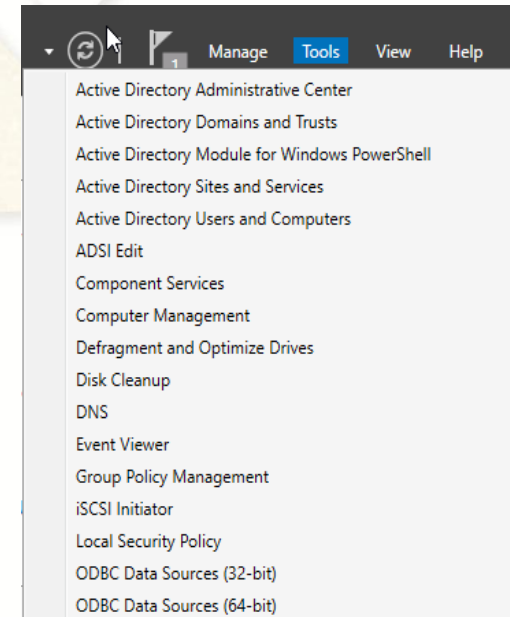
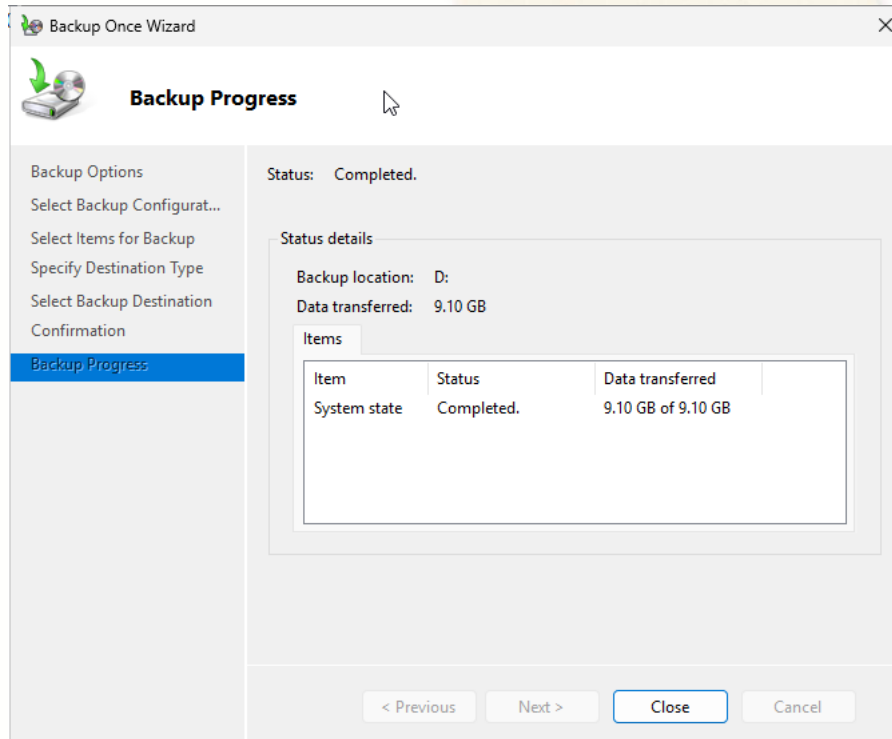
Backup Once AD (Cont.)

- Step 5: Select **Local Disk (D:)** or specific backup file location > Click **Next** > Click **Backup** button



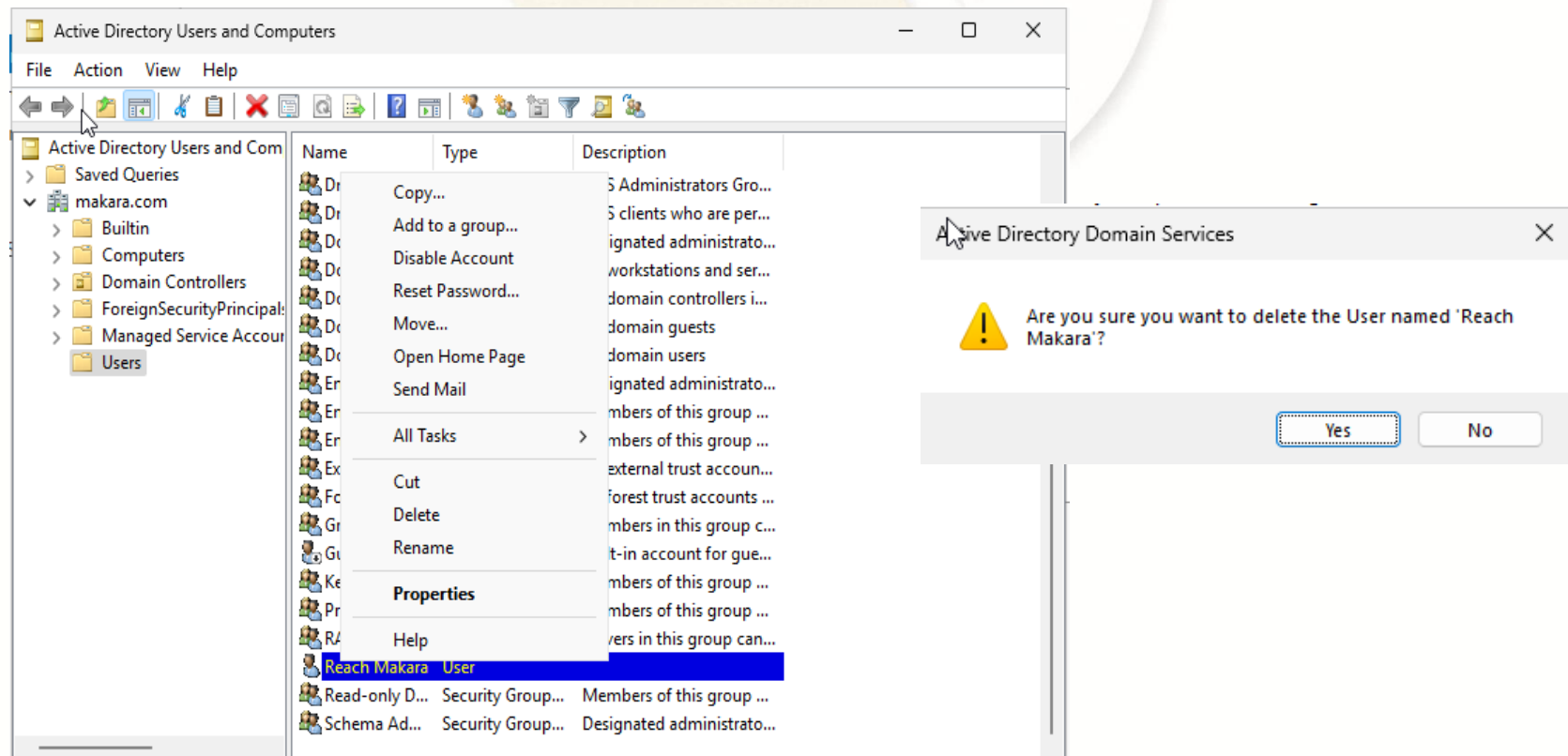
4.2. Backup Once AD (Cont.)

- Step 6: When **backup progress** has been **completed**, Click **Close** > Go to **Server Manager** and Click **Tools** menu > Click **Active Directory Users and Computers**



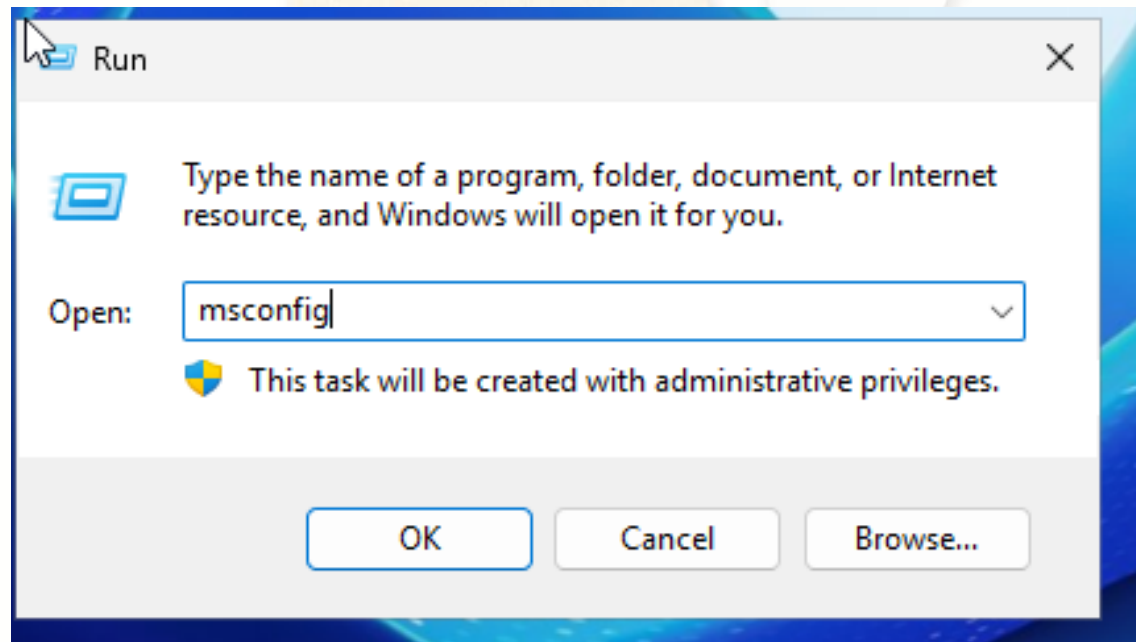
4.2. Backup Once AD (Cont.)

- Step 7: Right Click on a **user (Reach Makara)** > Select **Delete** > Click **Yes** button (**this user will restore after recovery process**)



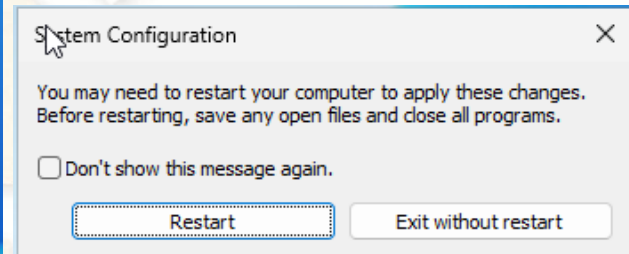
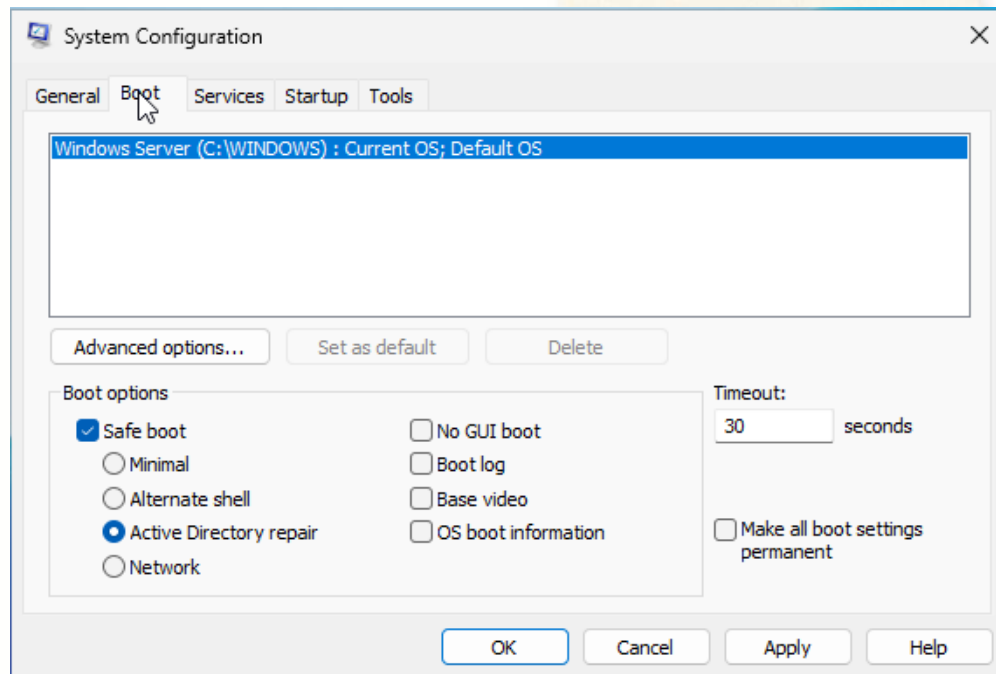
4.3. Restore AD

- Step 1: Go to **Run** by key **Windows + R** > Type **msconfig** and Click **OK**



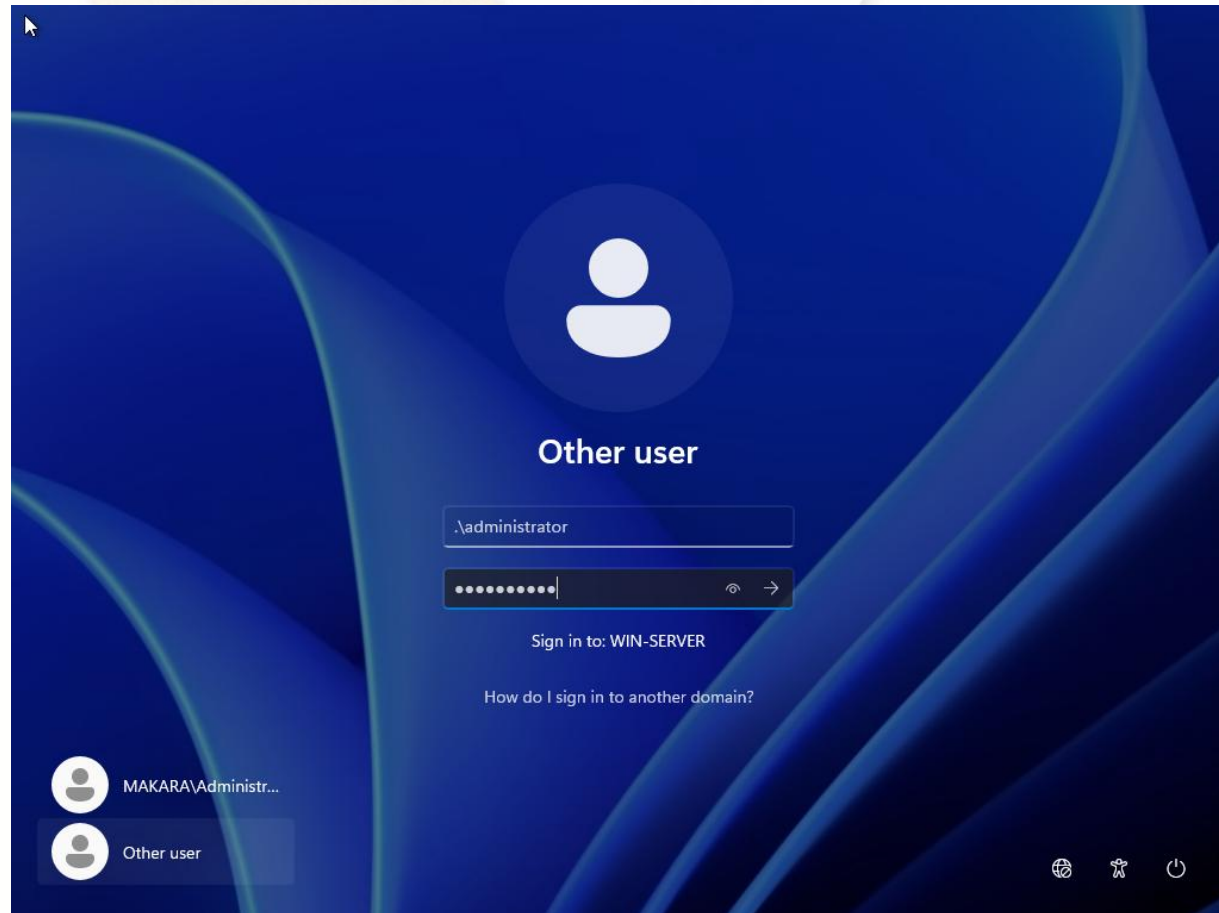
4.3. Restore AD (Cont.)

- Step 2: Select **Boot** tab > Check **Safe boot** > Select **Active Directory repair** > Click **OK** > Click **Restart**



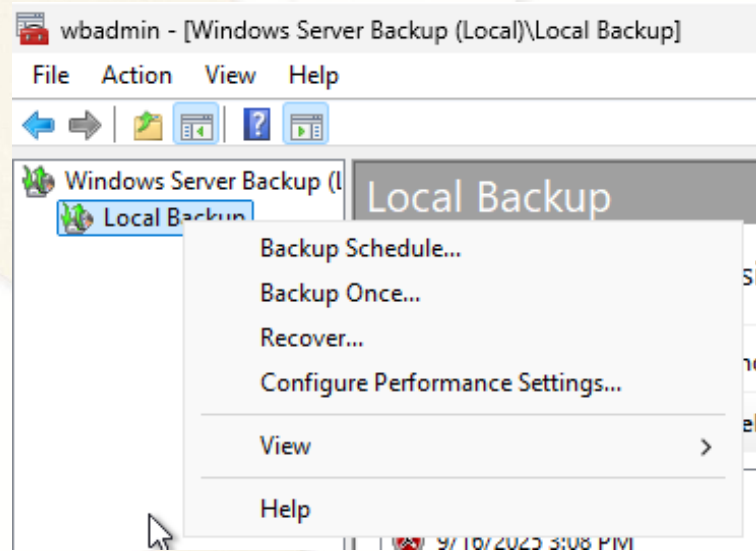
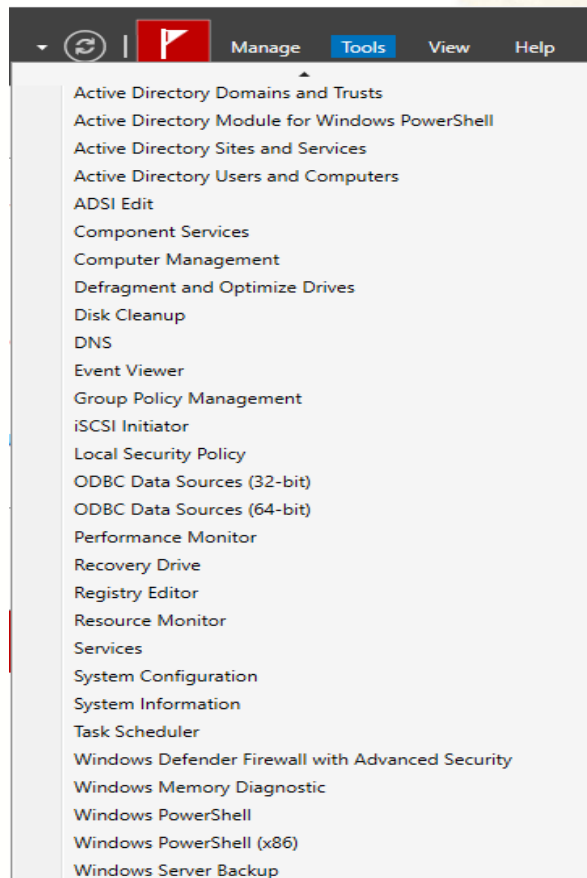
4.3. Restore AD (Cont.)

- Step 3: Click Other user > Enter **local account** information such as username (.\\Administrator) and password



4.3. Restore AD (Cont.)

- Step 4: Go to Server Manager and Click Tools menu
> Select Windows Server Backup > Right click on Local Backup > Select Recover...



4.3. Restore AD (Cont.)

- Step 5: Select **This server** > Click **Next** > Select recent **backup date** > Click **Next**

The screenshot shows the Windows Recovery Wizard with two windows. The background window is at the 'Getting Started' step, and the foreground window is at the 'Select Backup Date' step.

Recovery Wizard - Getting Started

You can use this wizard to recover files, applications, volumes, or the operating system from a backup that was created earlier.

Where is the backup stored that you want to use for the recovery?

☒ This server (WIN-SERVER)

☐ A backup stored on another location

To continue, click Next.

< Previous **Next >** Recover

Recovery Wizard - Select Backup Date

Getting Started
Select Backup Date
Select Recovery Type
Select Items to Recover
Specify Recovery Options
Confirmation
Recovery Progress

Oldest available backup: 9/18/2025 9:31 AM
Newest available backup: 9/18/2025 9:31 AM

Available backups
Select the date of a backup to use for recovery. Backups are available for dates shown in bold.

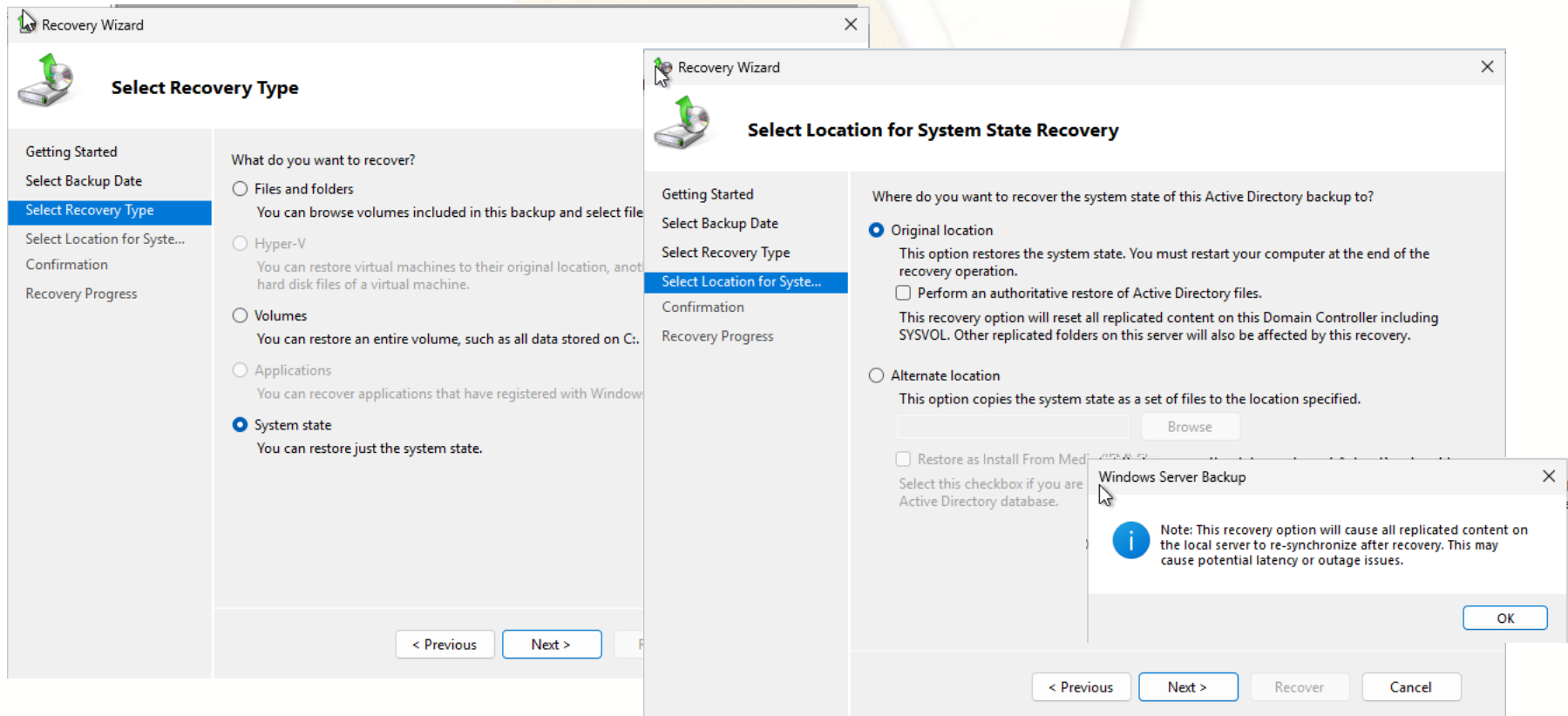
September 2025						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

Backup date: 9/18/2025
Time: 9:31 AM
Location: Local Disk (D:)
Status: Available online
Recoverable items: [System state](#)

< Previous **Next >** Recover Cancel

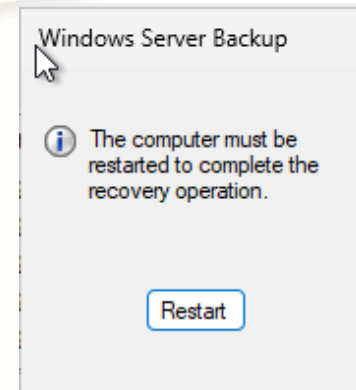
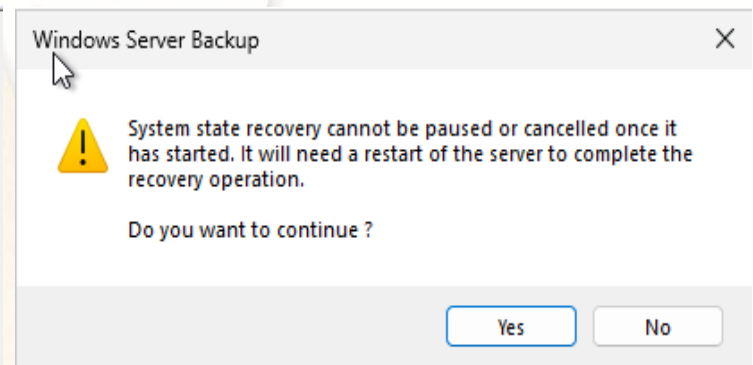
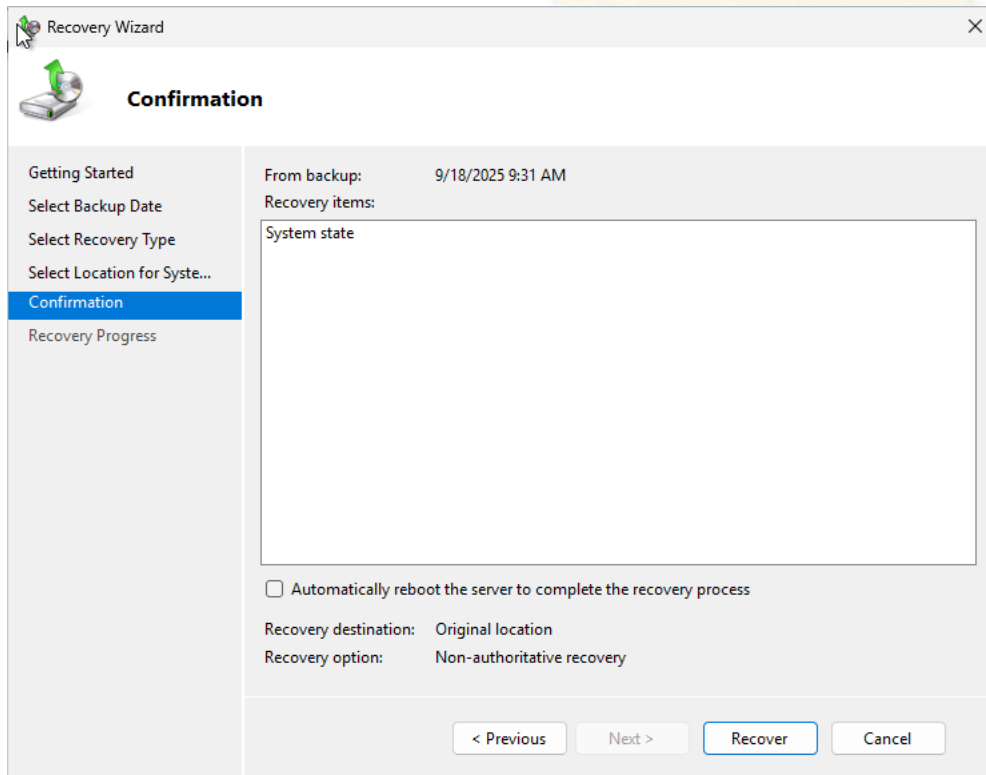
4.3. Restore AD (Cont.)

- Step 7: Select **System state** > Click **Next** > Select **Original location** > Click **Next** > Click **OK**



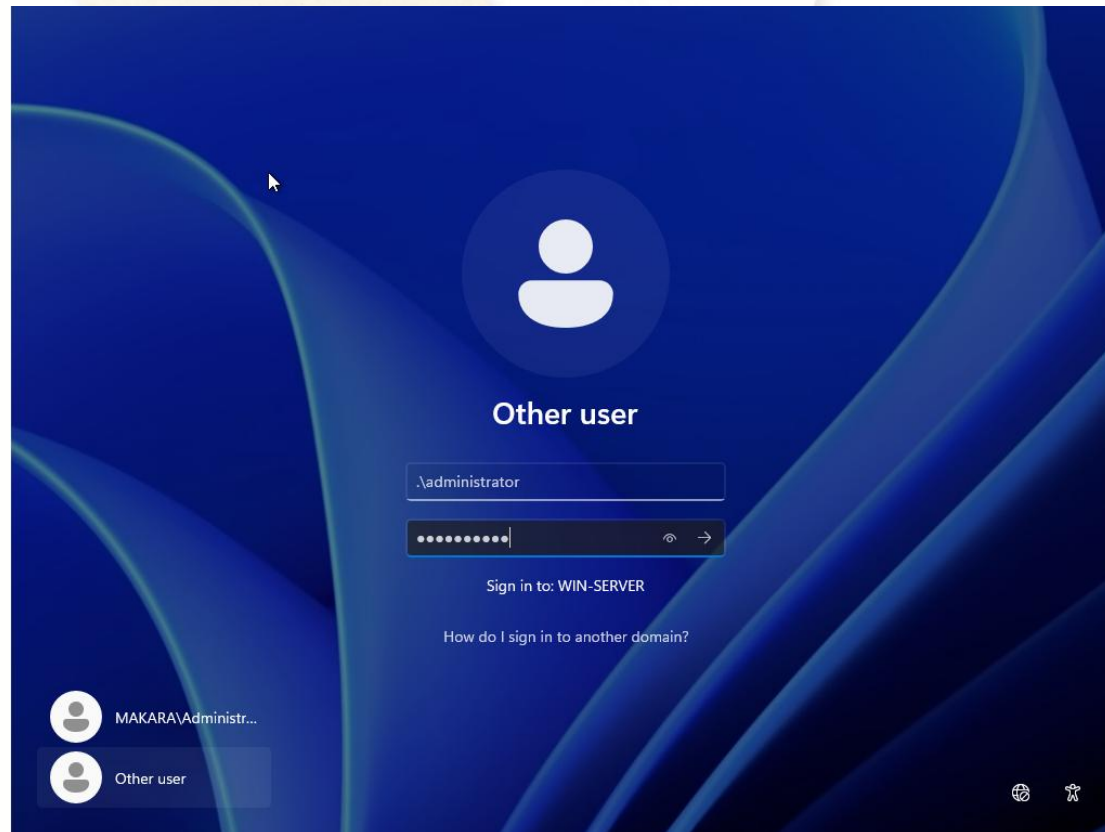
4.3. Restore AD (Cont.)

- Step 8: Click **Recover** > Click **Yes** button > Wait until **recovery process** has been **completed**, click **Restart** button



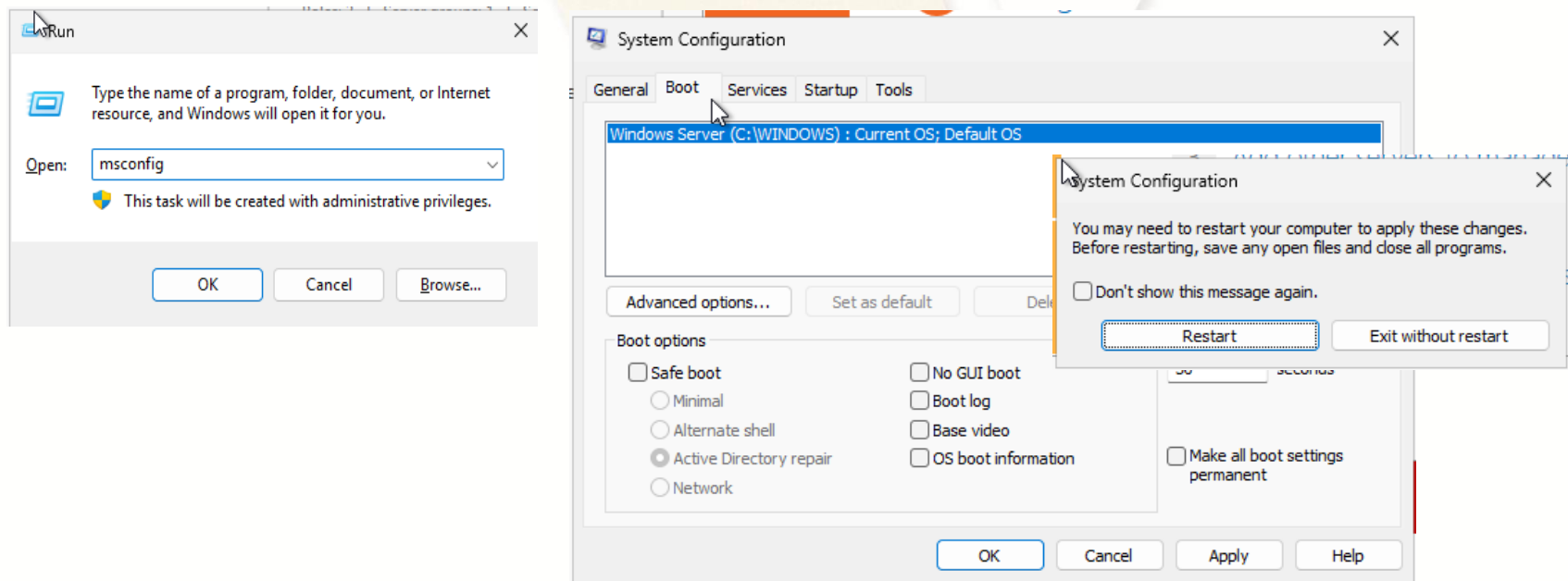
4.3. Restore AD (Cont.)

- Step 9: Click Other user > Enter **local account** information such as username (.\\Administrator) and password



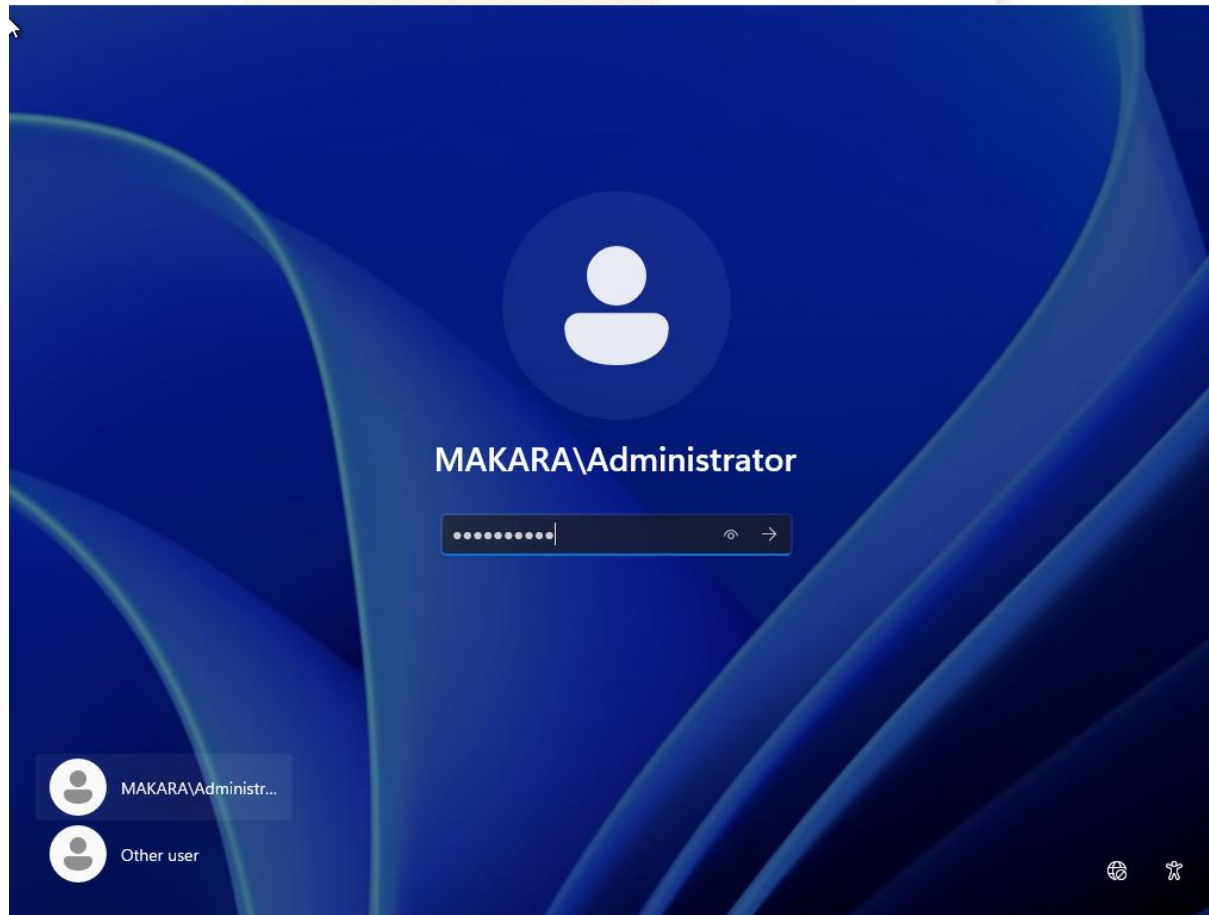
4.3. Restore AD (Cont.)

- Step 10: **Start + R** key to open **Run** Windows > Type **msconfig** and Click **OK** > Select **Boot** tab > Uncheck **Safe boot** > Click **OK** > Click **Restart**



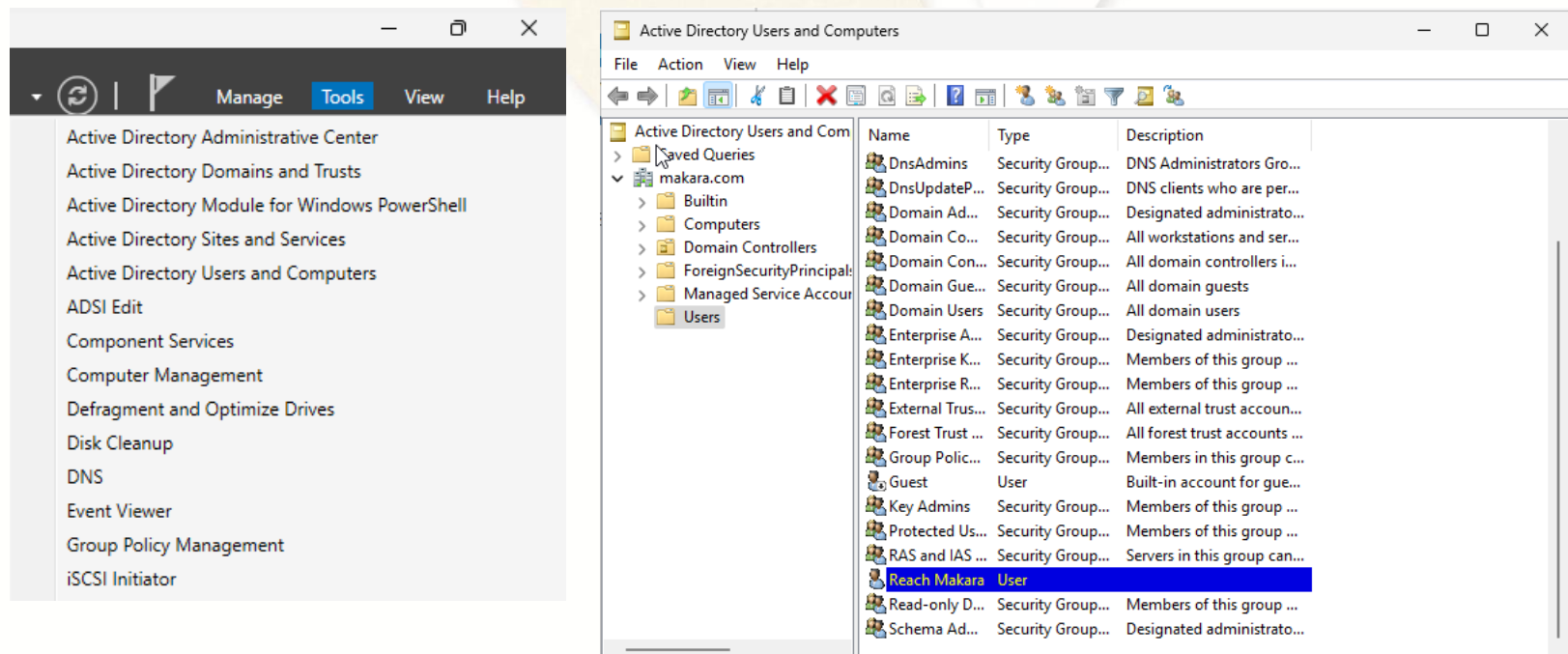
4.3. Restore AD (Cont.)

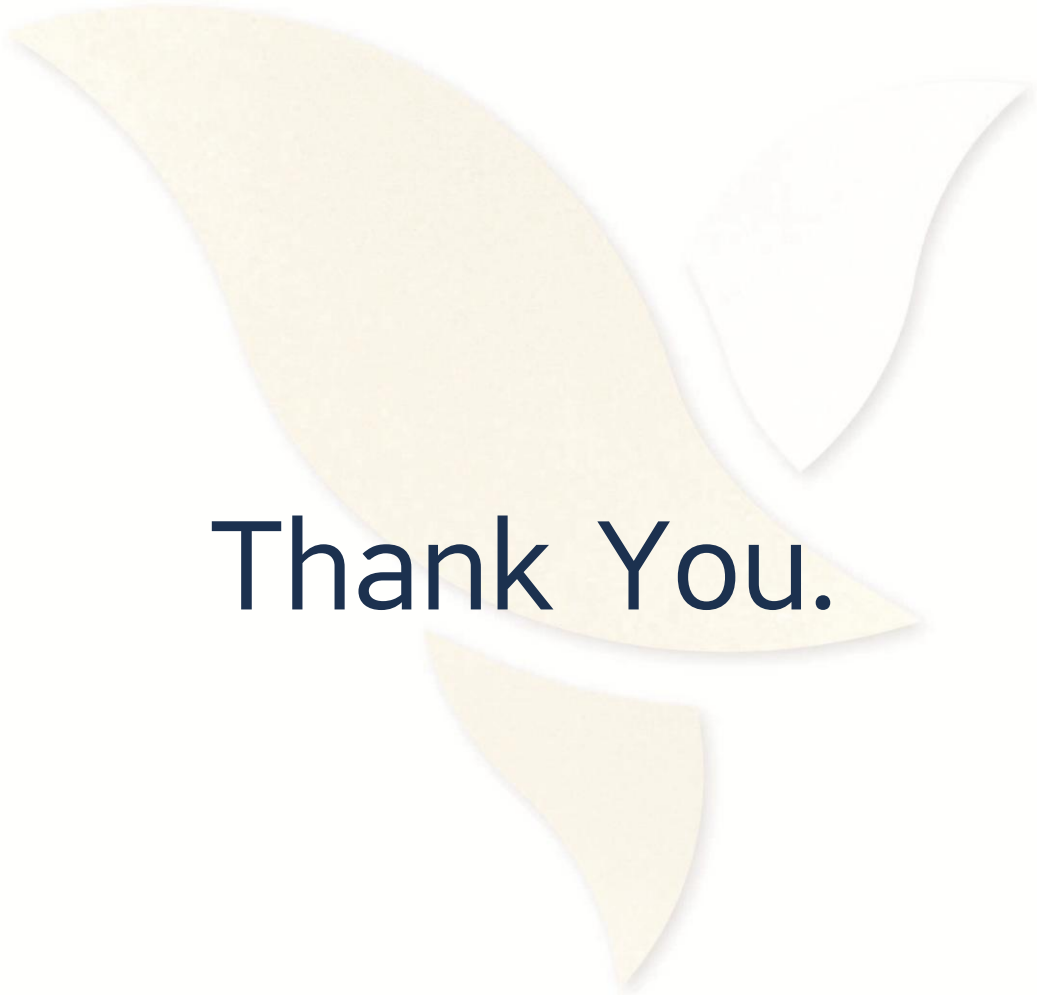
- Step 11: Click on MAKARA\Administrator > hit password and Enter for **Sign in**



4.3. Restore AD (Cont.)

- Step 12: Go to Server Manager and Click Tools tab > Click Active Directory Users and Computers > Go to Users Folder and in user list you will see **Reach Makara** has been **recovered**





Thank You.